Living with Density: The Impacts of California's By-Right Permitting for Accessory Dwelling Units*

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Abstract

In this paper we use a difference in discontinuities approach to study the impacts of discrete regulatory reforms for Accessory Dwelling Units (ADUs) on permitting activity, land values, and sale prices. This approach leverages the California Coastal Zone as a stringent and stable regulatory environment that changes discretely in space, while areas outside of the Coastal Zone receive regulatory relief. We find that ADU permitting approximately doubles after these reforms, although as a share of parcels at most 1% choose to build an ADU. However, these reforms do create an option value that capitalizes in the total sale price through increased land values. We estimate that the option value ranges from 5% to 10% of land values and is robust to a number of specification choices.

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1 Introduction

Economists, and to a growing extent, policymakers have linked local regulations on housing production to increased prices for housing services (Glaeser, Gyourko and Saks, 2006; Ganong and Shoag, 2017; Severen and Plantinga, 2018; Kahn, Vaughn and Zasloff, 2010; Bartik, Gupta and Milo, 2024).[1] However, addressing these barriers with policy reforms has proven politically challenging. Homeowners are often reluctant to support a policy that could negatively impact their home values, and localities are rarely eager to relinquish their discretion over the permitting and approval process for new housing (Gray, 2022; Dillon, 2019; Tobias, 2022; Garrick, 2021; Brinig and Garnett, 2013). As a result, "light touch density" has become a politically palatable approach to reform that focuses on loosening regulations for smaller units, such as accessory dwelling units (ADUs) (Pinto, Peter and Hamilton, 2022; Schuetz, 2025; Kahn and Furth, 2023). California has passed a series of bills since 2016 that de-regulate ADUs, which are small single unit additions to an existing housing structure.[2] Since 2016, permitting for ADUs has increased from a baseline of approximately 1,000 state-wide permits issued per year before 2016 to almost 40,000 permits issued in 2024 alone, as shown in Figure 1. This paper leverages a discrete spatial change in the California regulatory environment to isolate the impacts of these reforms.

Starting in 2016, the State of California began passing a series of pieces of legislation to reduce local discretion over approving ADUs. They made the permitting process by-right, where ADUs that met state-determined criteria *must* be approved within a set number of days. However, these regulatory reforms are superseded within the California Coastal Zone, which is an approximately 1 kilometer wide strip of land running up the entire coastline of California. Within this regulatory environment, the California Coastal Commission has control over land use through the coastal development permit (California Coastal Commission, 2022*a*,*b*; Severen and Plantinga, 2018; Kahn, Vaughn and Zasloff, 2010). Due to this discrete change in the regulatory environment, before 2016 parcels within the Coastal Zone needed a coastal development permit to construct an ADU and parcels outside of the Coastal Zone required a local permit. After 2016, parcels within the Coastal Zone still required a coastal development permit, but parcels outside of the Coastal Zone had their local permitting streamlined by state-wide preemption for ADUs. Our difference in discontinuities design will compare parcels just outside the Coastal Zone to parcels just inside the Coastal Zone before and after 2016 to estimate the impacts of the state's permitting reforms for ADUs (Grembi, Nannicini and Troiano, 2016).

Using our pre-registered difference in discontinuity approach, we investigate three hypotheses regarding the impacts of permitting reform (Cañedo Riedel and Hoxie, 2025). First, we hypothesized that permitting reform should increase the amount of permits issued discretely at the boundary. We find evidence that permitting for an ADU is about twice as common compared to permitting

[1] A large literature also connects these local distortions into the broader macro-economy (Hsieh and Moretti, 2019; Hoxie, Shoag and Veuger, 2023; Duranton and Puga, 2020, 2023).

[2] ADUs typically are under 1,200 square feet. They can be a stand-alone structure in a back yard, a converted garage, or at times a conversion of an existing part of the house. Sometimes ADUs are also referred to as casitas, in-law units, granny flats, secondary units, guest houses, cottages, among other names.

within the Coastal Zone.[3] Second, we hypothesized that ADU permitting reforms should increase land values for all residential parcels, even if they do not build an ADU. We find that residential parcels, the vast majority of which are zoned for single family homes, see an increase in land values per acre of between 5% and 10% after the reforms. We interpret this increase as an option value, as the reforms increased the highest economic use of the land and gave the owners the ability to build a potentially valuable ADU in the future. We demonstrate how this option can impact current prices in a simple net present value model in Subsection 6.3. Finally, we postulated that ADU reforms would likely result in a small and negative spillover onto the values of adjacent parcels. We find only suggestive evidence of negative spillovers using our design. This is likely because so few parcels actually exercise the option to build an ADU in our analysis sample.

This research contributes primarily to a large and active literature on housing and regulations. Several papers and textbooks link zoning rules to decreased housing supply and higher home prices (Glaeser, 2013; Brueckner, 2011; Glaeser, Gyourko and Saks, 2006; Bartik, Gupta and Milo, 2024; Ganong and Shoag, 2017; Baum-Snow and Han, 2024; Severen and Plantinga, 2018; Kahn, Vaughn and Zasloff, 2010). Moreover, Baum-Snow and Han (2024) find that about 40% of the supply response to price changes comes from small renovations, which could include ADUs. Our paper is an improvement over many papers that study local zoning since variation in local regulations tend to be endogenously chosen, whereas in our setting the state imposes the reforms on the localities (Gyourko and McCulloch, 2024; Bartik, Gupta and Milo, 2024). However, how effective decreasing local regulations are at lowering housing costs is currently being debated in the literature (Louie, Mondragon and Wieland, 2025; Duranton and Puga, 2023; Ganong and Shoag, 2017). Our findings help rationalize this divide. Although we find that permitting does increase as a result of reducing local discretion, we don't see a large shift in units as a share of all parcels. Therefore, it is unlikely that ADUs have meaningfully helped to depress rental prices in the broader market for small rentals, although they may reduce prices locally. However, the option value created by the ability to build an ADU does impact a meaningful number of parcels, and likely gets captured in aggregate measures of home prices commonly used in macro studies.[4]

Cross-plot spillovers have been a repeated theme in both the economics literature and policy debates. In some cases, large spillovers from wide-spread destruction can create incentives for urban renewal (Hornbeck and Kenniston, 2017). However, on a smaller scale meaningful cross-plot spillovers in prices have been difficult to see (Brooks et al., 2020). Negative spillovers can take the form of concerns that new development may decrease existing structure and land values by increasing use of local services, congesting local amenities (e.g. street parking), or diminishing

[3]Due to a difficulty with collecting pre-2018 permit data, we have temporarily used a regression discontinuity design to investigate the impacts of permitting, which is a deviation from our pre-analysis plan. However, we are working to collect the missing permits from local sources. Moreover, we hypothesized that Figure 1 suggests that we already have the vast majority of ADU permits issued since 2012.

[4]For example, Ganong and Shoag (2017) and Hoxie, Shoag and Veuger (2023) use housing values from the American Community Survey, and Louie, Mondragon and Wieland (2025) uses data from the S&P Cotality Case-Shiller US National Home Price Index and the American Community Survey. In these data sources land and structure values are not differentiated. However, it is also possible that increased housing production from reduced supply constraints could still be consistent with aggregate price increases in general equilibrium.

neighborhood aesthetics (Dillon and Khouri, 2016; Garrick, 2021; Gray, 2022; Brinig and Garnett, 2013). Although, there is limited evidence that ADU construction can result in negative spillovers within a close proximity to the new unit (Tanrisever, 2025; Davidoff, Pavlov and Somerville, 2022).[5] We find only suggestive evidence of small, negative spillovers from ADU permitting reforms.

We also provide the first causal evidence, to our knowledge, that loosening land use regulations can produce an option value. The idea of development as a type of option has long roots in economics Titman (1985); Riddiough (1997). Grout, Jaeger and Plantinga (2011) uses a regression discontinuity design to link Portland, Oregon's urban growth boundary to increases in land prices, but does not explicitly link these differences to an option value, but rather to a limited supply of developable urban land. The most related paper to our work is Cunningham (2007), which finds that growth controls around Seattle, WA reduced the real option value of land and accelerated development. Our findings suggest that the converse may also be true when regulations are relaxed.

In the following sections we will first discuss the landscape of reforms to ADU regulations in California. Then, we will describe our dataset and empirical approach. Then we will present results for permitting changes, land values, and spillovers. We then contextualize these findings with a discussion of robustness exercises and present a simple model of the option value to build an ADU. We end the paper with a few concluding remarks.

2 Housing Reforms in California

Since 1926 it has been clear that cities can have the power to regulate local housing construction through zoning (Village of Euclid v. Ambler Realty Co., 1926). However, the exact powers that cities possess have largely been determined by the states that created them (Davidson et al., 2020). In 2016, the State of California passed the first of several laws that aimed to encourage the construction of ADUs by removing local regulatory barriers enforced through zoning, permitting backlogs, parking requirements, and other hurdles (Dillon and Khouri, 2016; Minott, 2023; Gray, 2024).

Prior to 2016, building an ADU in a particular jurisdiction was largely under local control. Although the State passed legislation aimed at streamlining ADU permitting in 2002, local governments were still able to use the discretion of the permitting process as well as a myriad of additional requirements as a *de facto* veto of the state-wide reforms. Moreover, a second bill in 2003 was proposed to address local roadblocks, but Governor Schwarzenegger vetoed the bill since it encroached on local authority. This left cities with the power to impose additional site requirements and inspections, impact fees, parking requirements, and other cost-increasing measures on new ADUs (Brinig and Garnett, 2013).[6] Starting in 2016, the State has passed a series of reforms that stripped local jurisdictions of their abilities to impose barriers to ADU construction (Gray,

[5] Brueckner and Thomaz (2024) looks at changes in property values of units that elect to build an ADU in Los Angeles, and finds that ADUs increase property values by about 8%.

[6] For a more complete discussion of the regulatory environment in the 150 largest California cities surrounding ADUs, please see Brinig and Garnett (2013).

2024; Minott, 2023).

The first set of reforms came in the 2016 legislative session as Assembly Bill (AB) 2299 and Senate Bill (SB) 1069. These reforms required "by-right" permitting with a maximum process time of 120 days. This meant that if an ADU met existing zoning requirements, then the ADU must be approved. The reforms also required jurisdictions to allow ADUs on lots that already have single family homes on them; removed a number of burdensome requirements such as street access, new utility hookups, excessive setbacks; reduced existing parking requirements for ADUs near transit or built as part of a primary residence (Minott, 2023; County of Humboldt, 2017).

Continuing its efforts in the 2017 legislative session, the state passed SB 229 and AB 494. These bills clarified that local jurisdictions cannot ban homeowners from renting out ADUs separately from the attached residence, that ADUs can be part of proposed single-family housing projects, and limited parking requirements to only one additional space per unit. Then in 2019, the legislature continued its reforms to ADUs with three new bills, AB 68, AB 881, and SB 13.[7] Together, these pieces of legislation removed minimum lot sizes, capped setbacks, eliminated floor area ratios, prohibited owner-occupancy requirements, expanded garage conversions, and allowed ADUs in multi-family areas. These additional limitations on local governments were paired with increased enforcement tools through the California Department of Housing and Community Development. The state also limited the ability of HOAs to restrict ADU rentals in AB 670 (2019) and AB 3182 (2021), clarified that selling ADUs separately is legal in AB 587 (2019), and required permitting jurisdictions to communicate existing grants and financing programs to encourage ADU financing in AB 671 (2019). Finally, in 2022 the state clarified that all ADUs up to 16 feet high are permitted, limited the setback restrictions, and required localities to enumerate to applicants why their permits were denied (AB 2221 and SB 897) (Minott, 2023; Gray, 2024).

As a result of these reforms, a typical residential lot in California is entitled to build a 1,200 square foot ADU and a 500 square foot Junior ADU with only a 4 foot setback on the side and rear. Moreover, existing detached structures can be converted into ADUs of up to 1,200 square feet. These projects must be approved within 60 days (California Department of Housing & Community Development, 2026).

However, these reforms, generally, will not impact jurisdictions within the roughly 1 kilometer strip of land along California's coastline. The California Coastal Zone's land use has been governed by the Coastal Act and the California Coastal Commission since 1972 (California Coastal Commission, 2022*a*). The Coastal Act prohibits the state from imposing the reforms to ADUs that "increase the intensity of" land use, such as by-right permitting, reduced setbacks and other restrictions on local jurisdictions within the Coastal Zone (California Coastal Commission, 2022*b*). Instead, within the Coastal Zone, new development must receive a special coastal development permit, and any of the state-wide reforms may only be implemented by jurisdictions updating their local coastal plans voluntarily (California Coastal Commission, 2022*b*; Severen and Plantinga, 2018; Kahn, Vaughn and Zasloff, 2010).[8] In practice, the Coastal Zone is governed by local coastal plans,

[7] These bills also expanded the reforms to Junior ADUs, which are smaller than regular ADUs (Minott, 2023)

[8] As of this 2022 memo, only 5 cities have voluntarily adopted parts of the reforms to ADUs. These include Santa

which cover 131 segments spanning the state. The local jurisdiction (e.g. the City of San Diego, UC San Diego, or the County of San Diego) that overlaps with the Coastal Zone will file a plan that details how the land will be managed in accordance with the Coastal Act. These plans must be approved by the Coastal Commission, and can periodically be updated.

In Figure 2, we show a map of the San Diego County Coastline. We display the Coastal Zone, and its segments, in blue and incorporated cities in orange. These are superimposed on all real estate transactions in the Cotality (2025) data in 2016. Before 2016 and after 2016 in the Coastal Zone (the blue region), ADUs are subject to a stable, albeit stringent, regulatory environment. However, outside the Coastal Zone (in the orange and white regions) parcels benefit from the suite of reforms to ADUs enacted by the State legislature. This creates a spatial difference in discontinuities before and after 2016, where areas just outside the Coastal Zone will experience regulatory changes to ADUs while areas inside the Coastal Zone will not benefit from these reforms.[9] Critically, this stark change in the regulatory environment represented by the Coastal Zone boundary cuts directly through populous neighborhoods, which we (partially) demonstrate by showing 2016 real estate transactions as black dots in Figure 2. We will discuss the details of this research design in the following sections.

3 Data

To investigate the impacts of permitting reforms for accessory dwelling units we build a dataset that links property characteristics from assessment records, real estate sales data, and building permits. We construct this dataset for the 15 coastal counties of California from 2010 onward, as data are available.[10] Our final dataset contains just over 4 million residential parcels and over 2 million transactions between 2010 and 2025.

The Cotality (2025) property data create the backbone of our dataset. This data source collects and standardizes county assessor records which are used to levy property taxes. These records include information on assessed values for the structure, land, and total property value, building and lot characteristics, and location identifiers including coordinates and a street address.[11] Cotality (2025) also provides records of real property transactions, which include information on the real

Cruz, Santa Barbara, Encinitas, San Mateo, and Pacifica (California Coastal Commission, 2022*b*).

[9]There are more pieces of legislation that have been passed by the State of California to address housing affordability. These measures include the ADU reforms, as well as rezoning efforts surrounding transit dense housing, infill housing, and lot splitting among other reforms. To our knowledge, only the ADU reforms and the SB 35 environmental permitting waivers for infill housing. SB 35 does not modify local zoning laws, and only limits discretionary permit review. There have been about 160 SB 35 projects since its enactment in 2017. These projects are generally much larger infill projects, with 90% of SB 35 projects containing more than 10 units, and 74% of projects containing more than 50 units. This largely puts SB 35 projects farther away from the small residentially zoned lots that are the focus of this paper (Manji and Finnigan, 2023).

[10]From north to south, the 15 coastal counties of California are: Del Norte, Humboldt, Mendocino, Sonoma, Marin, San Francisco, San Mateo, Santa Cruz, Monterey, San Luis Obispo, Santa Barbara, Ventura, Los Angeles, Orange, and San Diego.

[11]In California, assessments are generally updated after real property changes hands or after construction, and year-to-year assessed values are capped to grow at no more than 2% each year (San Francisco Assessor Recorder's Office, 2025).

property being sold, the date of the sale, an arms-length transaction identifier created by Cotality, and a sale price in nominal dollars.

In addition to the property records from Cotality, we incorporate data on building permits from the State of California's housing element progress reports (California Department of Housing and Community Development, 2025).[12] As part of the post-2016 housing reforms, the State of California began requiring local jurisdictions (cities and counties) to report on the number of land use entitlements, building permits, and certificates of occupancy they issue each reporting year, by project, starting in 2018. This dataset is then linkable to our property and transaction data using either an assessor parcel number (APN) or by matching the reported addresses.[13]

We compare the state-wide totals of building permits from California Department of Housing and Community Development (2025) with previously available totals for ADU permits taken from Peterson (2021) in Figure 1.[14] The series for state-wide ADUs from 2012 to 2022 shows about 87,500 permits issued, with only about 4,000 permits issued between 2012 to 2016 (shown in solid blue). The more recent Housing Element tables provide more that state aggregates, they report permits from every jurisdiction at the project level since 2018 (the dashed orange line) (California Department of Housing and Community Development, 2025). The detailed data, from 2018 to 2022 alone account for at least 90% of all permits represented by the Peterson (2021) series.[15] Although we do not have any pre-reform permits to match, Figure 1 suggests that permitting before the reforms for ADUs was generally negligible.

Using the unique identifier for each parcel and provided latitude and longitudes from Cotality (2025), we calculate each parcel's distance to the California Coastal Zone boundary and nearest Coastal zone segment using shapefiles from California Coastal Commission (2014).[16]

With our matched permits, transaction, and assessment records in hand, we build two analysis datasets. For methods that will analyze the impacts of permitting reforms on transactions, we will use a dataset based on the Cotality (2025) transaction records, with building characteristics, assessment information, location, land use categories, and other parcel level information attached to these records using the unique property ID provided by Cotality. We then prune these data expanding on a process outlined in Frank et al. (2025). First, we limit our dataset to parcels that are ever listed as having a land use or property use that is for small residential uses.[17] Next,

[12] We are also in the process of collecting building permits from the local jurisdictions directly to supplement these records.

[13] To do this matching, we develop a hierarchical matching algorithm that prioritizes matching by APN and then uses the address information. For the address matching, we used Anthropic (2026)'s Claude AI to refine and improve our R code since a critical step was identifying address components from semi-structured address fields that can contain many locations. After we separate all of the listed addresses, we attach a permit to all addressess associated with it first using the APN, then an exact match on the address, then on the Google location key from the Google Maps Platform (2026), and finally on previous listed APNs. We describe this process in detail in Appendix **??**.

[14] This series comes from the California Department of Housing and Community Development, but we were unable to locate the original source. The same time series of state-wide ADU permits going back to 2012 is corroborated by Gray (2024) and Minott (2023).

[15] There are slight discrepancies between the two series. The detailed data from Tobias (2022) include unincorporated permits, and it is unclear whether or not these areas are included in the older series.

[16] To be included in our sample, a property must have a valid identifier and a coordinate.

[17] This includes parcels marked as single family residences or duplexes or with land uses marked as residential uses

we trim the sample to include only arms-length transactions, assessed values greater than $1,000 (real) USD, (real) sale prices between the 1st and 99th percentiles, plausible transaction volumes within a single year (99th percentile and above are trimmed), room to unit ratios in between the 1st and 99th percentiles, (real) land values per acre greater than $1 USD, and an internally consistent implied age of the structure based on the date of sale and the building date. We describe how these restrictions impact our final transaction sample in Table A.2.[18]

In order to evaluate the impacts of permitting reforms on the share of parcels that receive an ADU permit in a given year, we construct a balanced panel dataset. This dataset takes the Cotality (2025) assessment data and ensures that each unique parcel has one observation per year for 2010 to 2025. This requires us to summarize building and lot characteristics when a parcel has multiple assessments within a given year, as well as to expand our dataset to fill in gaps between assessments.[19] Additionally, we apply the relevant subset of restrictions for assessments. The parcel must have a small residential use, have an assessed value above $1,000 (real) USD, have a room to unit ratio outside the 1st and 99th percentiles, and have an internally consistent implied age. The impacts of each sample restriction are tracked in Table A.2.[20] The final parcel-year sample has just over 66 million observations between 2010 and 2025, although our permit sample restricts us to only use observations starting in 2018.[21]

We construct three primary outcome variables to capture changes in land values, total values, and permitting activity. Since land values are not measured directly in our transaction data, we combine the transaction sale price and the land share of total value from the first available assessment after the transaction.[22] We then take the total land price and divide it by the parcel land acreage to get the (real) land value per acre.[23] As shown in Table 1, on average, before the 2016 permitting reforms, transactions in the full 15 county sample $3.1 million per acre (2025 USD), compared to $4.5 million per acre (2025 USD) after the reforms.

To capture the impacts of spillovers, we use the (real) total sale price per acre, which combines the transaction sale price and the parcel land acreage from its assessment records. In Table 1, we can see that in generally total sale prices increased from on average $5.6 million per acre (2025 USD) before the reforms to almost $8 million per acre (2025 USD) between 2017 and 2025.[24]

Our third and final outcome variable is a dummy for whether or not a parcel received an ADU

between 1 and 10 units.

[18]We cannot use assessed alone to measure changes in land value as assessment growth in California is capped, and in most cases only updates when there is a transaction (San Francisco Assessor Recorder's Office, 2025).

[19]To expand the data, we take characteristics from the last available assessment.

[20]We explore how these sample restrictions interact with our research design in Appendix Table A.3. These restrictions differentially remove observations from inside the Coastal Zone, but these differences appear to be stable over time and therefore are not a threat to our difference in discontinuities framework.

[21]The vast majority, over 90% of our final dataset consists of single family homes, as shown in Table A.1.

[22]Typically, assessments are updated after arms-length transactions in California, so this is the best available measure of the land share of total value (San Francisco Assessor Recorder's Office, 2025).

[23]We deflate all monetary variables by the personal consumption expenditures chain-type price index to convert values to 2025 USD (U.S. Bureau of Economic Analysis, 2025).

[24]Note that if a 3,500 square foot home in La Jolla, CA is on a 0.2 acre (8,700 square foot) lot and sells for $5 million USD, then its price per acre is $25 million USD.

permit in a particular year. This variable takes a value of 1 if the parcel has an associated ADU permit, and a value of zero otherwise. We only use this variable in our panel dataset starting in 2018, as shown in Table 2. In the full county sample, only about 0.2% of parcels had an ADU permit in a given year. Although this is a flow measure, it still reflects the fact that ADU permitting, even with reforms, appears to be relatively rare. However, Table 2 does suggest that ADU permits are likely a large share of building permits overall.

We construct our running variable as each parcel's distance to the Coastal Zone boundary in meters. Values of the distance to the Coastal Zone (CZ) variable are less than 0 when a parcel is within the Coastal Zone, and greater than 0 when a parcel is outside the Coastal Zone, with the boundary itself taking a value of 0. We will use this running variable to define the bandwidth for our final analysis sample, as well as to assign parcels to distance bins from the coastal zone boundary.

In Tables 1 and 2, we present means and standard deviations for variables in our 1,000 meter bandwidth analysis sample. Columns 1 through 4 present summary statistics for the sub-sample of parcels inside the Coastal Zone from 2010 to 2016, inside the Coastal Zone from 2017 to 2025, outside the Coastal Zone from 2010 to 2016, and outside the Coastal Zone from 2017 to 2025. Across all Columns, around 90% of parcels are single family parcels. Buildings on these parcels tend to be relatively old, with build years around 1970.[25] The houses on these structures tend to have around 3 bedrooms, 2.5 bathrooms, and have a building square footage of about 2,000 square feet.[26]

We graphically show the location of our final 1,000 meter bandwidth sample in Figure 3. We present the density of our Coastal Zone sample by latitude as a solid blue line. Our sample ranges from San Diego in the south, just north of 35 degrees, to the Oregon border at almost 42 degrees. The majority of our analysis sample is located in Southern California, with a large Coastal sample in both San Diego and Los Angeles. As we move north, there is a meaningful sample around Santa Cruz and San Francisco, and then a final set of parcels in Eureka. The density for the remaining (non-Coastal) parts of the 15 counties are shown as an orange dashed line. Even though the ocean-adjacent parts of California include some of the most desirable real estate in the United States, the wide range of this sample will allow us to evaluate the impacts of ADUs in a number of different communities. We will present our methodology for doing so in the following section.

[25] Effective build year is the year in which the parcel had its currently assessed features constructed. So, if a new extension to the house was added, such as an ADU, the effective build year would update.

[26] Full sample statistics, including means, standard deviations, minima, and maxima can be found for the full county sample in Table A.1. It is important to note that the sample is mostly, but not entirely, single family units. We intentionally include non-apartment or condo multi-family parcels as these types of units would not have their highest economic use increased by the ability to build an ADU. However, given that we have a dataset of tax parcel, rather than lots, it is the case that several commonly owned units may be combined in a single parcel, which can skew the extremes of the distribution for characteristics such as bedrooms, bathrooms, etc.

4 Methods

Our primary hypothesis is that the enactment of by-right permitting for areas outside of the Coastal Zone will increase the flow of new ADUs building permits authorized compared to the static regulatory environment within the Coastal Zone.[27] To evaluate these hypotheses, we will use a spatial difference in discontinuities design similar to Grembi, Nannicini and Troiano (2016). This strategy combines the flexible control polynomial of a spatial regression discontinuity, while leveraging discrete changes in the regulatory environment across the Coastal Zone boundary. In addition to our main hypothesis, we will also evaluate several secondary hypotheses. First, it may be the case that allowing ADUs may raise all total land values outside the coastal zone as the option value associated with the *ability* to build an ADU may increase the value of lots, even if they do not exercise it. Second, we can also evaluate whether or not building an ADU is associated with significant cross-plot spillovers, which may be either positive or negative.

To evaluate our primary hypotheses that reforms increased permitting. We will estimate the following specification for 2018 to 2025 using our balanced parcel dataset:[28]

$$Y_{ict} = \alpha + \beta_1 OutCZ_i + \beta_2 f(DistCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict} \tag{1}$$

where Y_{ict} it is an indicator variable taking a value of 1 if parcel i in county c received and ADU permit in year t. $OutCZ_i$ is an indicator taking a value of 0 if lot i is within the Coastal Zone, and therefore has stable regulations, or a value of 1 if lot i is outside the Coastal Zone and therefore will receive the state-wide permitting reform package in 2016. $f(DistCZ_i)$ is a polynomial that is a function of lot i's distance from the closest point on the Coastal Zone boundary.[29] In order to control for pre-reform characteristics, we construct a set of hedonic fixed effects using pre-2016 parcel characteristics, $\theta_{(i)}$. These fixed effects are a set of interacted bins that capture the parcel's build year, land use category, number of bedrooms, number of bathrooms, and building square footage before the permitting reforms.[30] We also include county (λ_c) and time (τ_t) fixed effects. The county fixed effects will control for constant characteristics of county assessors offices as well

[27] These hypotheses, as well as our methodology, have been pre-registered on OSF before beginning any of the data work for this project, as required by our grant from Arnold Ventures (Cañedo Riedel and Hoxie, 2025). Given that we have not finished our permit collection efforts, we will deviate from our stated plan by necessity when handling our permit data.

[28] This specification deviates from our pre-analysis plan since we are in the process of collecting the pre-2018 building permits. Even so, it is in our opinion that including these additional permit data will be unlikely to overturn our results as the vast majority of permits for ADUs are included in the current version of the paper according to Figure 1.

[29] We will use first degree and second degree polynomials, each interacted with the $OutCZ$ dummy to allow for slope changes across the boundary.

[30] We construct fixed effects in a similar style to Yagan (2019), which include the full interaction of the following groups, with bins for missing values in each category. The bins are: above or below county median effective age (and a missing bin), land use category (single family, other, or missing), number of bedrooms (0-2, 3-4, 5 and up, and missing), number of bathrooms (0-2, 2-4, 4 and up, and missing), and within county tertiles of building square footage (and a missing bin).

as other static characteristics of the county. The time fixed effects will control for factors impacting all lots in a given period, such as interest rate changes. In some specifications, we replace the time fixed effect with a border-segment by year fixed effect to ensure that all comparisons are within local segments of the Coastal Zone boundary. ε_{ict} is an error term. We will cluster standard errors at the parcel level as the Coastal Zone boundary is plausibly exogenous to any individual parcel, as argued by Severen and Plantinga (2018) and Kahn, Vaughn and Zasloff (2010). We will present estimates for parcels within 1,000 meters of the Coastal Zone boundary.

The main assumptions for our spatial regression discontinuity model is that permitting activity across the Coastal Zone boundary would be the same absent the difference in the regulatory environments. Explicitly, we are assuming that lots inside the Coastal Zone boundary and lots just outside the Coastal Zone boundary would have built ADUs at similar rates to the pre-period differences if there was no ADU reform package passing in 2016. Since Figure 1 suggests that pre-reform permitting activity was negligible, we will proceed with assuming that pre-reform ADU permitting activity was essentially zero on both sides of the boundary. Additionally, we are assuming that the regulatory environment jumps sharply at the Coastal Zone boundary, whereas relevant features for building ADUs besides regulation change smoothly. Severen and Plantinga (2018) and Kahn, Vaughn and Zasloff (2010) both show evidence in support of this claim. Critically, we are identifying the effects of a package of reforms, as we will be unable to separate the impacts of by-right permitting from the effects of eliminating parking requirements and minimum lot sizes for ADUs.

In order to detect the effects of ADU reforms on the option value of land, we will use the same specification as in Severen and Plantinga (2018) which uses transaction sale prices as the outcome variable. We estimate the following equation using or transaction sample from 2010 to 2025:

$$\begin{aligned} ln(Y)_{ict} =& \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3 f(DistCZ_i) \\ &+ \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict} \end{aligned} \tag{2}$$

where $ln(Y)_{ict}$ is the log of the real land value per acre for parcel i in county c sold in year t. In this specification we will cluster standard errors at the parcel level.[31] Just as with Equation 1, we can replace the time fixed effects with segment by time fixed effects that keep the comparisons local to each segment of the Coastal Zone boundary. Since we also have multiple transactions for a large subset of parcels, we present a repeat sale model that includes a parcel fixed effect that controls for static characteristics of the parcel. This requires us to limit our sample to parcels sold at least once both before and after 2016. We present these specification for our 1,000 meter from the boundary bandwidth.

This difference in discontinuity design, developed by Grembi, Nannicini and Troiano (2016), compares changes in land values on parcels just inside the Coastal Zone boundary to parcels just outside the Coastal Zone boundary before and after the reform. The resulting estimator relies on a similar logic to difference in differences and regression discontinuity designs. The core assumption

[31] Since this specification is not limited to repeat sales, these standard errors will in practice be similar to those in White (1980).

is that, local to the cutoff, land values would have evolved similarly in the absence of the post-2016 reforms to ADUs. However, unlike the regression discontinuity design, stable differences in the characteristics of parcels across the Coastal Zone boundary are accounted for through differencing.

To detect spillovers local to the boundary, we will use a more flexible spatial regression difference in discontinuities following the design in Hornbeck and Kenniston (2017) to capture spillovers. Parcels just inside the Coastal Zone should be unable to take advantage of the new ADU law, but they will be exposed to any spillovers associated with the new construction. We can estimate a flexible specification as follows for 2010 to 2025:

$$ln(Y)_{ictb} = \alpha + \sum_{b} \beta_b (I\{Bin = b\}_i \cdot Post_t) + \theta'_{(i)}\gamma + \phi_b + \tau_t + \varepsilon_{ictb} \tag{3}$$

where $ln(Y)_{ictb}$ is the log of the real total sale price (land and structures) per acre on parcel i, in county c, in year t, and in Coastal Zone distance bin b. Here, we will construct a series of dummies for 50 meter coastal zone distance bins, $b \in \{1, .., B\}$, with an omitted bin 500 meters within the Coastal Zone as a comparison group.[32] Our set of estimates, β_b, will capture how prices evolve across a set of bins relative to the comparison bin. If spillovers are present, then prices should change just on the inside of the coastal zone, reflecting either positive or negative spillovers from the permitting reforms outside the Coastal Zone.

Finally, as of 2021, there are 5 jurisdictions that overlap with the Coastal Zone that updated their Local Coastal Programs to better accommodate aspects of the state-wide ADU reforms, even though they could not be forced to do so by the state. These areas are Santa Cruz, Santa Barbara, Encinitas, San Mateo, and Pacifica, which are all moderately sized cities (California Coastal Commission, 2022*b*). Even though these cities have updated their ADU requirements, it is unclear what meaningful changes were made in practice to allow new ADUs. In future iterations of this work we will present specifications that drop segments of the border that overlap with cities within the Coastal Zone that update their ADU permitting.

5 Results

We will investigate three primary impacts of the post-2016 set of permitting reforms in this paper. First, we will investigate the reform package's direct impact on permitting of ADUs. Permitting activity reflects directly how the legal changes are impacting the costs of building an ADU. However, even in the absence of pulling a permit and building an ADU, by lowering the costs of building an ADU the reform package may have created an option value that is capitalized in the land. Finally, we will investigate if there are noticeable spillovers onto nearby parcels just inside the Coastal Zone that may be exposed to the negative effects of the ADU law, but won't be able to construct and ADU themselves. We can express these effects in terms of the total price change:

[32] Our selection of the 450 to 500 meters bin as the reference bin is arbitrary. We picked it because the average distance from the boundary in the Coastal Zone is close to this bin, as shown in Table 1.

$$\begin{aligned}\Delta TotalPrice =& \Delta LandValue + \Delta StructureValue \\ &+ \Delta LocationSpillovers + \Delta StructureSpillovers\end{aligned} \tag{4}$$

Where the change in the total price (per acre) can be decomposed into four main piece. The first term, $\Delta LandValue$, reflects changes in the economic value of the land parcel. This term captures the option value created by ADU reforms, which may increase the highest potential economic use of impacted parcels. Critically, this option value can arise even without a significant increase in permitting activity, since the option value reflects what *could* go on the land and generate either a stream of utility value or rents for the owner (or a future owner).[33] The second term, $\Delta StructureValue$, changes with the value of the built structures on the parcel, which would also include the direct effects of adding an ADU to the property and any rental income generated by the ADU.

The second half of Equation 4 captures spillovers on to the parcel in two forms. Spillovers can be associated with the location of the parcel, $\Delta LocationSpillovers$, or with the price effects of nearby units on the market value of the structure, $\Delta StructureSpillovers$. The location spillovers typically get a lot of focus in zoning debates, as these can take the form of congestion forces that lower the total price of the property. These can include decreased on-street parking, increased use of local services like school districts, or just negative perceived neighborhood change.[34] However, these spillovers can also increase total prices by increasing neighborhood amenities. This might arise if ADUs also encourage new residents, which may attract shops and other businesses.[35] Finally, if ADUs increase the amount of rental units in an area, then the price for rental units may fall, generating negative spillovers to the value of the unit, $\Delta StructureSpillovers$.

5.1 Permitting

The first measure of the impact reforms to ADU permitting we investigate is changes in permitting activity. First, we will explore these differences in permitting across the Coastal Zone boundary graphically in Figure 4. We display permits in bins by their distance from the Coastal Zone boundary, where negative values (in orange) show locations within the Coastal zone that did not receive permitting reforms and positive values (in blue) show locations that were impacted by the state permitting reforms. Each bin covers 50 meters and the Coastal Zone boundary is marked by a line at 0. In Sub-Figure 4a, the vertical axis shows counts of permits issued per parcel[36]. We can see that in terms of volume, ADU permitting activity more than significantly increases when moving from the stringent Coastal Zone to the recently reformed area just outside of the Coastal

[33] For a simple net present value model of option values in real estate, see McMillen and O'Sullivan (2013).

[34] For some examples, see Gray (2022), Garrick (2021), and Dillon (2019).

[35] Both Severen and Plantinga (2018) and Kahn, Vaughn and Zasloff (2010) view the Coastal Zone as creating positive neighborhood amenities by restricting development and other congestion forces.

[36] We will also produce permits per parcel figures.

Zone.[37] In Sub-Figure 4b, we also show how ADU permitting changes as a share of all building permits. In the more lenient non-Coastal Zone areas ADU permitting is about 65% of all building permits issued, whereas inside the Coastal Zone ADU permitting is only about 45% of all permits issued. In both measures, the sharp difference in the regulatory environment is directly reflected in permitting activity.

In Table 3, we use our balanced parcel dataset and regression discontinuity design to show how permitting reforms outside the Coastal Zone increased permitting.[38] Our outcome variable is a dummy taking a value of 1 is a parcel in a given year has any ADU permit associated with it, and 0 otherwise. In Column 1, we include our baseline specification which only includes the $1st$ degree polynomial, county fixed effects, and year fixed effects. Our coefficients suggest an increase in ADU permitting of about 50% when comparing parcels just outside the Coastal Zone to parcels just inside the Coastal Zone. Cumulatively, over the 7 years represented by these data, almost 1% of parcels outside the coastal zone will get an ADU permit that wouldn't have absent these reforms. In subsequent columns, we add additional polynomials of the running variable, segment by time fixed effects, and hedonic fixed effects based on the pre-reform characteristics of the parcels. Our results appear to be robust to these alternate specifications, including our preferred specification in Column 5, which uses a first degree polynomial, county fixed effects, and segment by time fixed effects.

These results do suggest a meaningful change in the regulatory environment occurs when moving from the Coastal Zone to the rest of the state. However, if we were to take the direct change in permitting activity as the only impacts of the post-2016 permitting reforms, then we would have to conclude that these impacts were relatively limited as a very small fraction of parcels, so far, have elected to build an ADU.

5.2 Land Values

For smaller residential lots, the ability to add an ADU meaningfully increases the highest economic use of that lot since the ADU can generate both rent and value for the current or future owners. This option value associated with the ability to build an ADU should be present regardless of the amount of permitting activity and should be capitalized in the land value. We explore how the post-2016 permitting reforms impacted land values using our transaction dataset and difference in discontinuities design in Table 4.[39]

Column 1 of Table 4 shows our baseline specification for the impact of permitting reforms on land values. This column only uses a first degree polynomial in distance to the Coastal Zone boundary, a county fixed effect, and a year fixed effect. Our estimate suggests that permitting reforms raised land values by about 10% after 2016. Across the additional columns, we try alternate specifications using second degree polynomials of our running variable, replacing the year fixed effect with the

[37] In terms of raw counts of permits, activity more than doubles.
[38] This table follows Equation 1.
[39] This table follows equation 2 above.

segment by time fixed effect to keep comparisons local, and adding in a set of static hedonic fixed effects. We show our preferred specification in Column 5, which includes a first degree polynomial, county fixed effects, and segment by time fixed effects. These inclusions erode our estimates slightly. We find that permitting reform, using our preferred specification, increased land values by about 7.6%. These findings appear to be quite robust, with all models presented in Table 4 showing land value increases between 5% and 10% that are all statistically significant at least at the 1% level.[40]

We can show this change in land values visually in Figure 5. The graph shows 50 meter bins of transactions by their distance from the Coastal Zone boundary between 2010 and 2016 (orange squares) and between 2017 and 2025 (blue circles). The vertical axis shows the log of the real land value per acre. The effect of the permitting reforms can be seen by comparing the gap between the pre-2016 (orange) and post-2016 (blue) observations just before and just after crossing the Coastal Zone boundary. The gap widens in the figure after crossing the Coastal Zone boundary, visually corroborating our change in land values associated with the option of building an ADU found in Table 4.

Although the permitting reforms may not have put an ADU in every back yard, it did increase the value of these back yards, which could have significant fiscal impacts through property taxes. Property taxes in California are artificially low as a result of Proposition 13 as taxable value updates after a sale, when the property is re-assessed (San Francisco Assessor Recorder's Office, 2025). Even so, if the land values across the state have increased, as those properties are sold jurisdictions that rely on the property tax will see a fiscal gain.[41]

5.3 Spillovers

The public concern when discussing zoning is typically a discussion about spillovers. My neighbor may use her property in ways that negatively effect my property. This could be by taking up street parking, increasing utilization of local services, building something ugly, or simply competing down rental rates that I can get for my property. Motivated by Equation 4, we show how total sale prices, which include both land and structure values, change on a per acre basis as a result of the permitting reforms.

We present our estimates of Equation 3 in Figure 6.[42] This figure presents coefficients that capture how the (log of) real total sale prices per acre changed before and after the 2016 reform legislation. All coefficients are relative differences from changes in a reference bin, which includes transactions $[-500, -450)$ meters inside the Coastal Zone, which is right around the average distance for the transaction sample inside the Coastal Zone according to Table 1. The horizontal axis shows the bin's (median) distance from the Coastal Zone boundary, with negative values representing

[40] These results are consistent with previous findings from Ihlanfeldt (2007), who finds that increased land use regulations increase structure prices and decrease land prices. Our findings are symmetric, finding that decreased land use regulations increase land prices.

[41] This also makes a back of the envelope calculation of the total property tax benefit non-trivial.

[42] This figure shows estimates from a version of Equation 3 that includes a first degree polynomial, county fixed effects, and segment by time fixed effects.

transactions inside the Coastal Zone that did not receive the permitting reforms after 2016. If spillovers are present, they should be shown in the coefficients just inside the Coastal Zone.

In Figure 6 we find only modest evidence of spillovers from permitting reforms across parcels into the Coastal Zone. The four estimates just inside the coastal zone ($[-200, 0)$) are modestly negative, showing at most a spillover of 2%, are not significant at traditional confidence levels. By contrast, a strong jump in estimates can be seen after crossing the Coastal Zone boundary, reflecting the land option of around 7% shown in Table 4. These results suggest that the option value likely dominates concerns over negative spillovers in this setting.

6 Discussion

6.1 Robustness

In addition to the robustness presented in our main tables, we provide several additional exercises to further probe our results. First, we explore how bandwidth selection impacts our results for both estimates of Equations 1 and 2. We present these alternative choices in Figure A.1. In Sub-Figure A.1a we show estimates for the coefficient on exposure to permitting reforms, $OUtCZ$, following Column 1 from Table 3. These estimates include county fixed effects and year fixed effects. We present a range of alternative bandwidths from 50 meters to 2,000 meters. We find that, in general, our results become larger as we approach the boundary, but our standard errors increase as our sample size is diminished.

In Sub-Figure A.1b we explore alternative bandwidths for our estimates for the effects of changes in land values from Table 4. We also show our baseline estimates from Column 1, which only include county fixed effects and a year fixed effects. Coefficients for $OutCZ \cdot Post$ get larger in magnitude as our bandwidth limits the sample to transactions closer to the Coastal Zone boundary. Moreover, as the sample size decreases our standard errors increase. However, for all bandwidths in both Sub-Figures A.1a and A.1b, our results are consistently statistically significant and qualitatively similar to our preferred specification and preferred bandwidth of 1,000 meters.

Next, we probe the land value results in Table 4 in two additional ways. In Appendix Table A.4 we limit our sample to only parcels that had a transaction both between 2010 and 2016 and between 2017 and 2025. This allows us to include a parcel fixed effect, which will control for any static characteristics of the parcel that we may be unable to capture in our main specifications. Comfortingly, our estimates are similar to those presented in Table 4 and statistically significant at the 1% level, albeit slightly attenuated.

Finally, we explore trying to isolate changes in land values using updating hedonics in Appendix Table A.5. We estimate a modified version of Equation 2 where we use the log of the real total sale price per acre on the left hand side and also allow the binned hedonic fixed effects to update as characteristics of the structures on the property change over time. This alternative approach again produces qualitatively similar estimates to those in Table 4 that are all significant at the 1% level.

6.2 Heterogeneity

There may be substantial heterogeneity across Coastal Zone segments that may make some areas more likely to take up ADU permits or benefit from a broader set of land uses than others. We explore these differences by running Equations 1 and 2 segment by segment. Although the segments have different numbers of parcels in them, our overall treatment effects are a weighted average of the treatment effects in each segment.

In Figure 7, we show estimates of Equation 1 with county fixed effects and year fixed effect for each Coastal Zone segment.[43] The figure reveals substantial heterogeneity in the impacts of permitting reforms on permitting activity for ADUs. Areas with high estimated treatment effects likely have characteristics that make them more amenable to ADUs, which previous research has linked to middle income and Hispanic communities (Marantz, Elmendorf and Kim, 2023). Broadly, we find similar patterns. More Hispanic and Middle income communities have some of our strongest estimates for permitting changes across the Coastal Zone boundary. These areas include the largely Hispanic Chula Vista, National City, Imperial Beach, and Otay Mesa in southern San Diego County, the agricultural community of Guadalupe, the middle income urban area of Pacific Beach in San Diego, Venice Beach and the LAX area in Los Angeles to list a few segments with large permitting differences. These areas likely are near employment opportunities for workers who may benefit having access to ADUs as a smaller housing option.

In Figure 8, we estimate Equation 2 with county and year fixed effects for each Coastal Zone segment individually.[44] Here, we find that many of the same areas that saw additional permitting have larger land value increases. These areas include Otay Mesa, National City, parts of the City of San Diego, Chula Vista. However, there are also some different segments that appear to have high land option values without large jumps in permitting. These include Crescent City in Del Norte County, Hermosa Beach in Los Angeles, Carlesbad in Northern San Diego County, and Pacific Grove in Monterey County. These areas appear to include more affluent areas, where uses such as short term rentals may be especially relevant.

6.3 The Option Value of an ADU

We propose a simple option model to help understand our results. We use a net present value framework, which is common in real estate, in a similar style of model to Severen and Plantinga (2018) and McMillen and O'Sullivan (2013). Here, the utility for a parcel is given by the equation:

$$U_t = s + lap_t \tag{5}$$

where, s is the annual rents generated by the existing structure and land, l is the parcel's suitability for an ADU, which we will call simply quality, a is an indicator taking a value of 1 if the parcel

[43] These specifications are analogous to Column 1 in Table 3.
[44] These estimates are analogous to Column 1 of Table 4.

builds an ADU, and p_t is the time-specific (net) market price for the ADU.[45]

There are two time periods in this model in which the owner can elect to put an ADU on her parcel. The market prices for the ADU will take a known value of p_1 in the first period, and then it will realize either a high or low value in period two and forever after following $p_t = \rho p_H + (1-\rho)p_L$. Assume that $p_L < 0 < p_1 < p_H$. For simplicity, the choice of building an ADU is not reversible and owners face a penalty if they build and ADU in the low-price state of the world that scales with their lot quality, l. ADUs can only be built in period one or period two, for simplicity. The lot then accrues rents given its structure composition.

First, let's consider the case in which an owner builds an ADU on her parcel in period one, before she knows the long-term price realization. We can write this as a disconted flow of utilities:

$$V_Y^1 = s + alp_1 + \sum_{t=1}^{\infty} \frac{s + alE(p_t)}{(1+i)^t} - c \tag{6}$$

Here, we discount using a known interest rate i. Building an ADU incurs the net cost of c, which can include the cost of materials and labor used for both the structure and the permitting process. The reforms considered in this paper, in general, reduce the one-time cost of building an ADU, c.

Next, we consider the case in which the owner doesn't build an ADU, which can be written as:

$$V_N^2 = \sum_{t=0}^{\infty} \frac{s}{(1+i)^t} \tag{7}$$

where the owner simply enjoys the amenity value of her back yard and non-ADU structures, regardless of the price of an ADU.

In period 2, the owner is able to observe the realized price, p_t, that will persist indefinitely before choosing to build her ADU. Since we have assumed that $p_l < 0$, we know the owner will only elect to build an ADU in the event of the high draw for ADU values. We can write the value of building an ADU in the second period as:

$$V_Y^2|p_H = \sum_{t=0}^{\infty} \frac{s}{(1+i)^t} + \sum_{t=1}^{\infty} \frac{alp_H}{(1+i)^t} - \frac{c}{1+i} \tag{8}$$

When the owner waits and builds an ADU in period 2 after realizing high prices, she enjoys the land amenity as before and the persistent streams of ADU rents. By deferring building by one period, the cost of building is also discounted by a period. However, under certain cost-conditions, the owner will never elect to build an ADU.

$$V_Y^2|p_H - V_N^2 = \sum_{t=1}^{\infty} \frac{alp_H}{(1+i)^t} - \frac{c}{1+i} \tag{9}$$

Which is less than zero when:

[45] The price of an ADU could be thought of as net of any lost amenity values in the land. The ADU price could also reflect owner utility or market rents.

$$c > \sum_{t=1}^{\infty} \frac{alp_H}{(1+i)^{t-1}} \tag{10}$$

An owner will never build an ADU when the cost to build exceeds the total future revenue streams generated by the ADU. We will call this $\bar{c}$. We interpret the suite of ADU reforms as effectively lowering c such that $c < \bar{c}$ for a greater number of parcels.

If $c < \bar{c}$, the value of waiting to build an ADU in period two is a weighted average of the value of building an ADU given the high prices, and the value of never building an ADU. We can write this as:

$$\begin{aligned} EV_W &= \begin{cases} \rho(V_Y^2|p_H) + (1-\rho)(V_N^2), c < \bar{c} \\ V_N^2, c \geq \bar{c} \end{cases} \\ &= \begin{cases} \sum_{t=0}^{\infty} \frac{s}{(1+i)^t} + \rho[\sum_{t=1}^{\infty} \frac{alp_H}{(1+i)^t} - \frac{c}{1+i}], c < \bar{c} \\ \sum_{t=0}^{\infty} \frac{s}{(1+i)^t}, c \geq \bar{c} \end{cases} \end{aligned} \tag{11}$$

We can then think about the value of waiting to build an ADU in period 2 instead of building an ADU in period 1, before the prices are realized (when $c < \bar{c}$). We can write this option value as:

$$EV_W - V_Y^1 = c[1 - \frac{\rho}{1+i}] - al[p_1 + (1-\rho)\sum_{t=1}^{\infty} \frac{p_L}{(1+i)^t}] \tag{12}$$

where the first term is the difference in build costs between building in period 1 and period 2 and the second term is the stream of payments if the low state for ADU prices is realized, representing a loss to the owner. The value of waiting exceeds the value of building an ADU in period 1 if:

$$c[1 - \frac{\rho}{1+i}] > al[p_1 + (1-\rho)\sum_{t=1}^{\infty} \frac{p_L}{(1+i)^t}] \tag{13}$$

Therefore, it is better to wait to build an ADU if the value of the difference between the expected build costs from deferring building is greater than the rents from the ADU in the low price state of the world. Notice that as the costs of building an ADU, c, decreases, then the likelihood of deferring an ADU should decrease as well.[46] Similarly, the option value decreases when the probability of realizing high rents for ADU, ρ, is higher.

Finally, the option value is a function of the parcels' lot quality, l. This value could reflect the parcel's desirability in terms of neighborhood, location to jobs, or other land-specific amenities. We can write the derivative of the value of waiting with respect to land amenities as follows:

$$\frac{\partial EV_W - V_Y^1}{\partial l} = -p_1 + (\rho - 1)\sum_{t=1}^{\infty} \frac{p_L}{(1+i)^t} \tag{14}$$

[46] Notice that $[1 - \frac{\rho}{(1+i)}] > 0$, indicating the derivative of Equation 12 with respect to c is positive.

The option value is increasing in lot quality when the risk adjusted losses from building an ADU when low rents materialize are less than the negative baseline rent in period 1:

$$(1-\rho)\sum_{t=1}^{\infty}\frac{p_L}{(1+i)^t} < -p_1 \tag{15}$$

Finally, we can derive the option value with respect to the probability of realizing high prices for ADUs.

$$\frac{\partial EV_W - V_Y^1}{\partial \rho} = \sum_{t=1}^{\infty}\frac{alp_L}{(1+i)^t} - \frac{c}{(1+i)} \tag{16}$$

Since the deflated costs and losses from building an ADU when low rents materialize are both negative, the option value is decreasing in ρ. This means, as the likelihood of realizing high rents for ADUs increases, the option values diminish.

Solving this model numerically, we can show these patterns in Figure 9.[47] For a given level of lot quality for ADUs, l, total (net) cost of building c, and probability of high rents ρ, we show which parcels will elect to either build an ADU today in period 1 (dark blue), wait to decide to build an ADU in period 2 (orange), and never build an ADU (light blue). In the first panel, when high rents are certain, there is no option value and parcels either build an ADU right away or never build, due to low lot quality and prohibitive building costs. When the probability of high future rents, $\rho = 0.5$, decreases in panel 2, a wedge of middle cost and high lot quality parcels find it beneficial to wait to decide to build an ADU, reflecting a positive option value. When we decrease the probability of high rents even further, $\rho = 0.1$, even more lots decide to wait before building.

This simple model illuminates a few key elements of our results. First, a sufficiently high cost of building an ADU, $c > \bar{c}$, can make building an ADU unattractive regardless of the realization of rents, p_t. Permitting reforms, which lower the cost of building an ADU as part of c, should increase the number of ADUs and the option value of being able to build an ADU if the reforms move $c < \bar{c}$. Second, when the value of the ADU is uncertain, whether from rental market conditions or other factors, an option value can arise since there is value in the ability to gain additional information about future market conditions. This option value arises when development is delayed. The exact share of lots that see increases in option value versus that build immediately after reform depends on the lot suitability for ADUs, the costs of building, the probability of high rents to recoup the cost of production, and the possible downside risk from low rents. Third, given that many variables, which may differ across space, impact the decision to build an ADU, wait, or never build we believe that our results likely reflect an environment in which costs to build an ADU are around \$200 per square foot for many parcels, and that the likelihood of high rents in the future are less than

[47] The model is solved for 10 years. Interest raters are set at 10% (i) and rents from existing structures (s) are set at \$50,000 per year. We use a 1,200 square foot ADU that rents for \$1,200 per month in the base period, \$2,400 with high rents, and -\$1,200 with low rents for a base land quality l. We calculate solutions for all combinations of $l \in \{0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2\}$, $c \in \{100, 150, 200, 250, 300, 350, 400, 450, 500\}$ with c in thousands of USD, and $\rho \in \{0.1, 0.5, 1\}$. These build costs reflect industry ranges from Henson (2023).

certain, possibly even below 50% given that so few parcels elect to build an ADU.

7 Conclusion

In this paper we have shown that permitting reforms in California increased the volumes of permits issued for ADUs. However, given the low volume of permitting overall, the law's principal effects are most likely captured by an option value associated with the *ability* to build an ADU. This option value is capitalized into the sale price through increased land values. We find that permitting reforms increased land values by between 5 and 10%. These findings are robust to a range of model specification choices. This option value likely reflects why ADU reforms have been so politically palatable. The ADUs themselves do not appear to be associated with large and negative cross-plot spillovers. However, they allow the property owner to increase the utility of her own property and raise the value of the property even if she doesn't exercise the option to build an ADU.

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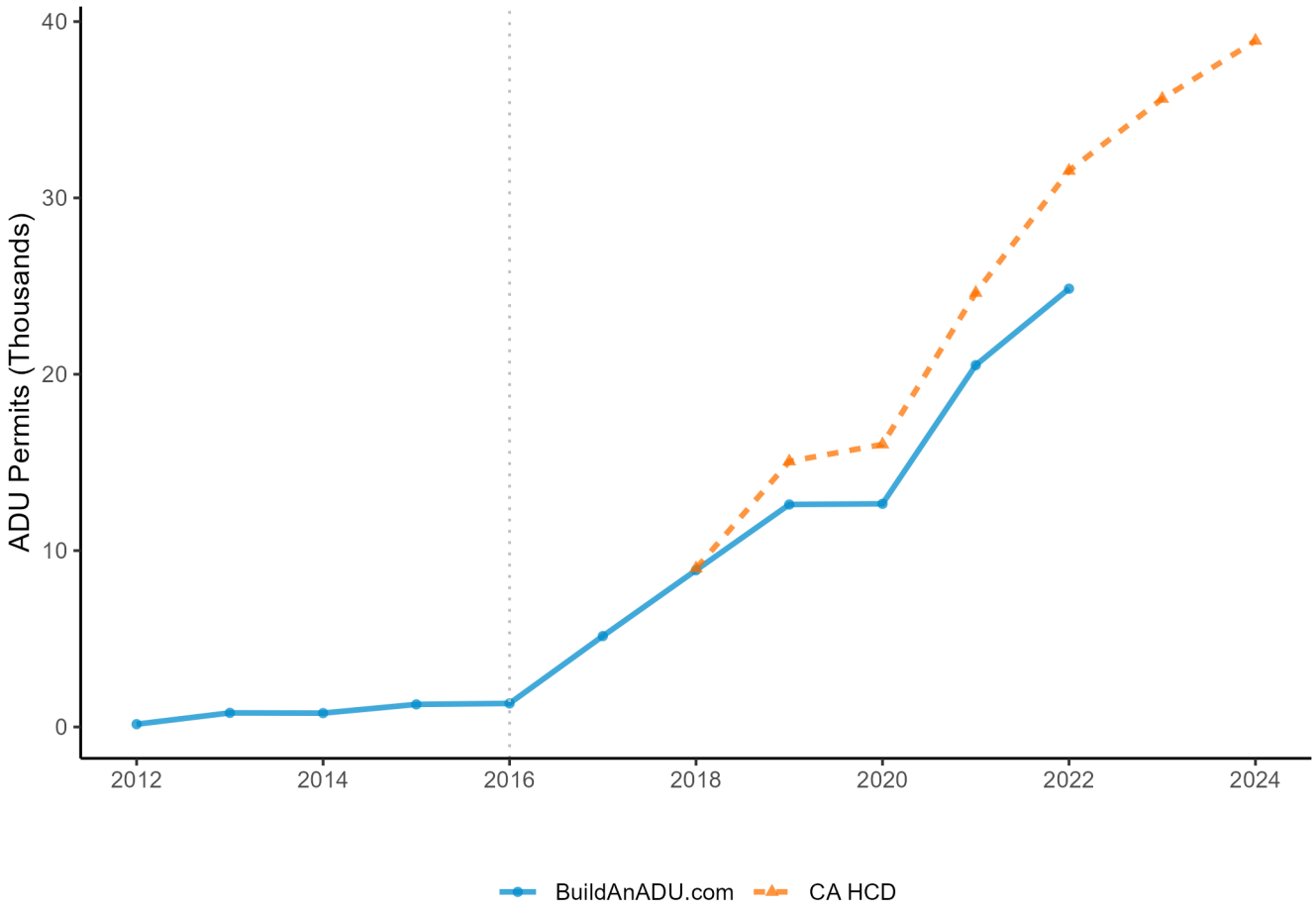
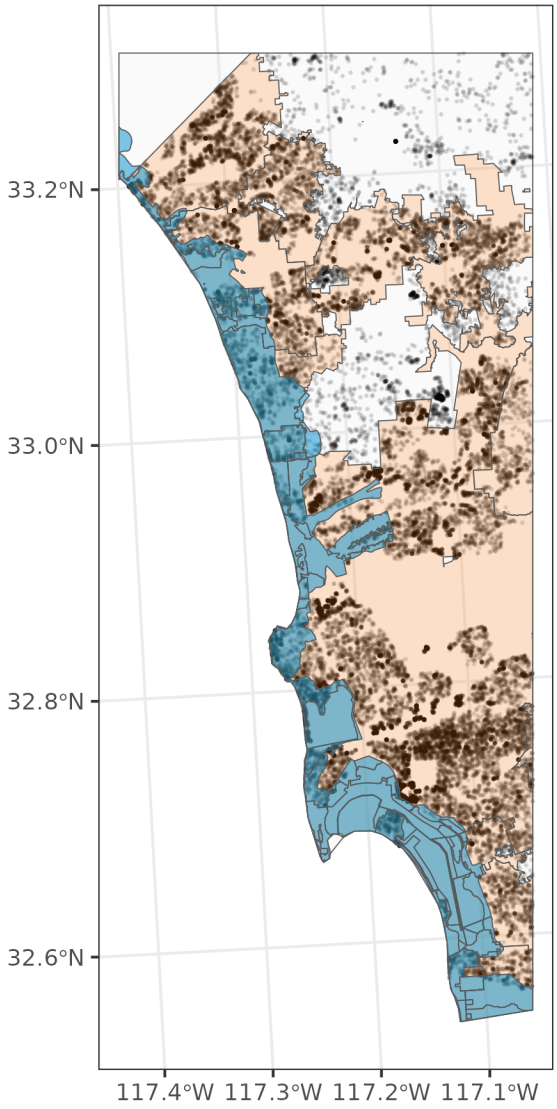
Figure 1: State-Wide Accessory Dwelling Unit Permits Issued in California from 2012 to 2024



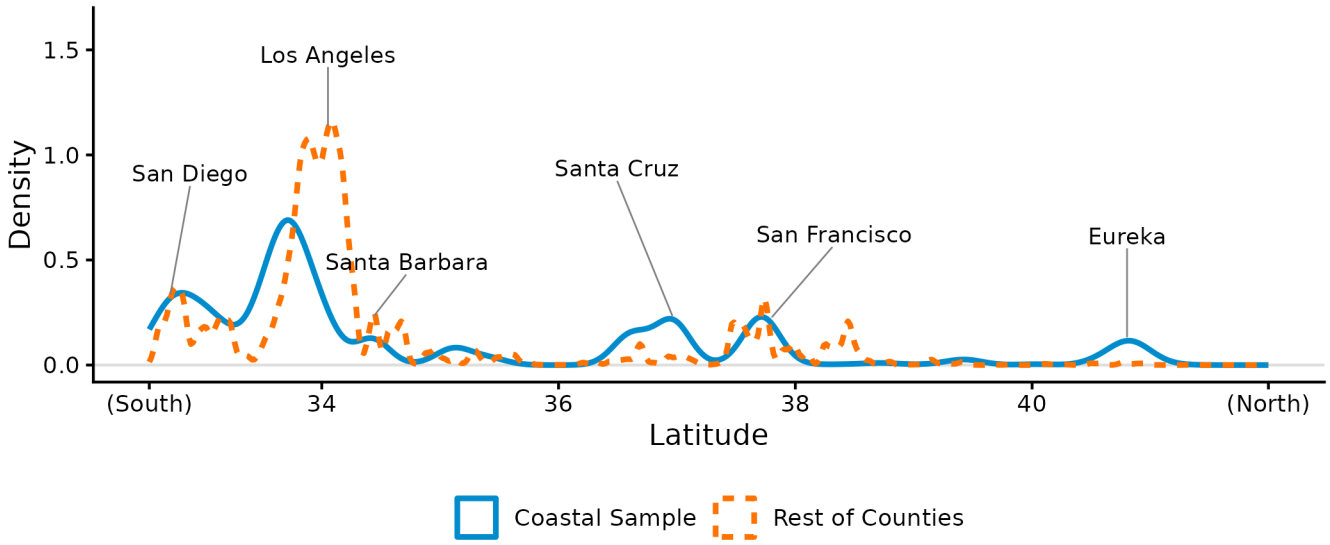
This figure shows a time series of accessory dwelling unit permits issued in the state of California from 2012 to 2024. The figure uses two data sources. The blue line shows permits counted by Peterson (2021), which collects data from the California Department for Housing and Community Development. The orange and dotted line shows permits according to the California Housing Element Progress Report (Table A2) data (California Department of Housing and Community Development, 2025).

Figure 2: Map of San Diego Coastal Zone, Incorporated Cities, and Real Estate Transactions in 2016



This figure shows the California Coastal Zone in San Diego County. The Coastal Zone is shown in blue. The orange area shows incorporated cities near the coast, and the black dots show all real estate transactions in 2016 from the Cotality (2025) data.

Figure 3: Density of Cotality (2025) Sample Parcels by Latitude

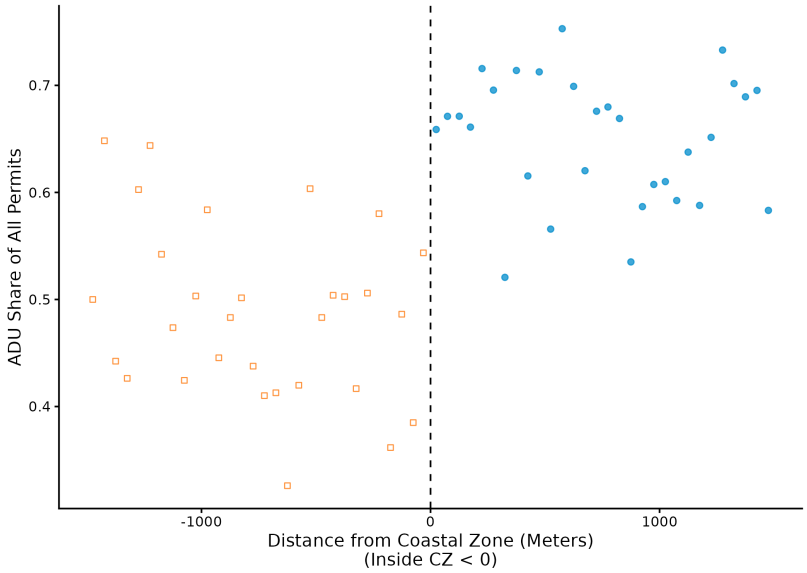


This figure shows the density of the Cotality (2025) property sample by latitude. The figure shows unique residential parcels from the year 2016. The solid blue line shows the sample within 1,000 meters of the Coastal Zone boundary, and the dashed orange line shows the remaining sample in 15 coastal counties outside of 1,000 meters from the Coastal Zone boundary. The horizontal axis shows parcel latitude, with Mexico abutting the southern portion of the sample on the left hand side, and Oregon abutting the northern portion of the sample on the right hand side. Selected cities are marked for reference.

Figure 4: Accessory Dwelling Unit Permit Activity Around the Coastal Zone Boundary after 2018

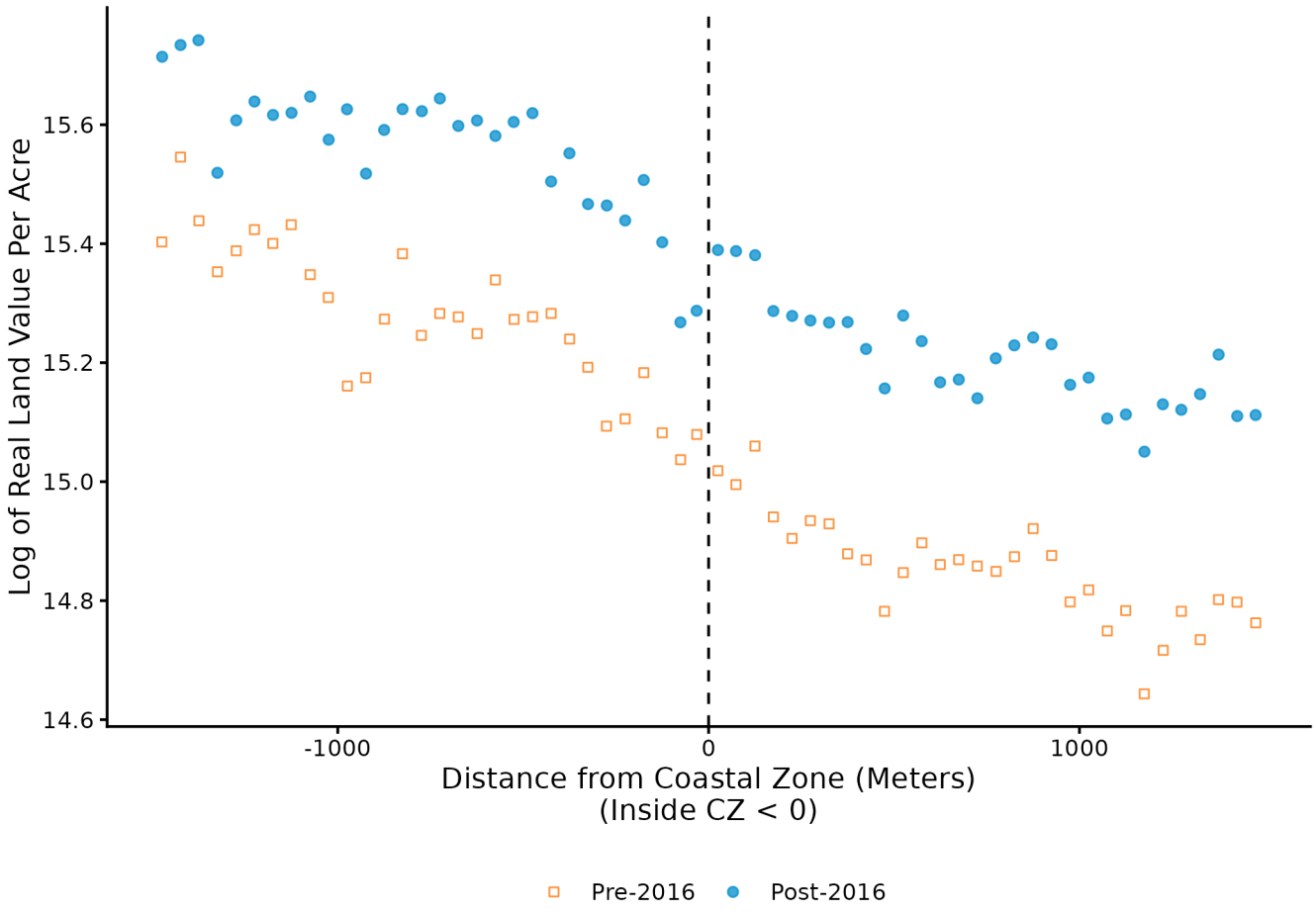
(a) Panel A: Accessory Dwelling Unit Permits Per Parcel by Distance Bin to the Coastal Zone

(b) Panel B: Accessory Dwelling Unit Permits as a Share of All Building Permits by Distance Bin to the Coastal Zone



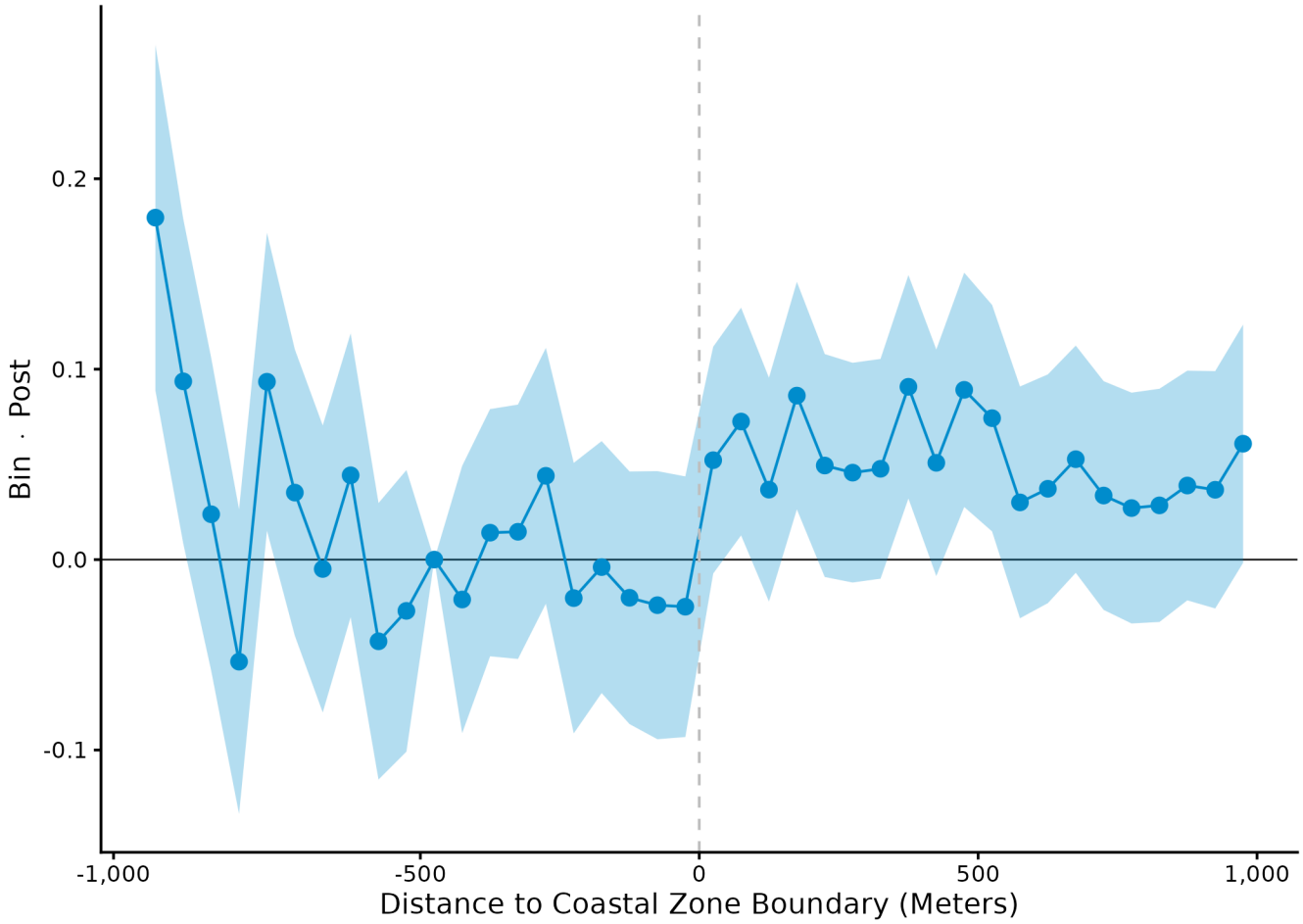
This figure uses data from Cotality (2025), California Department of Housing and Community Development (2025), and California Coastal Commission (2014) to show how permitting activity for ADUs changes across the Coastal Zone boundary between 2018 and 2025. The horizontal axis in both panels shows each bin's distance from the Coastal Zone boundary, where negative values represent parcels inside the Coastal Zone that do not receive the permitting reforms (orange squares). We group parcels in 50 meter bins. In Panel A, we show the number of ADU permits on a per-parcel basis for each bin. In Panel B, we scale the permit counts by the total number of building permits issued in each bin.

Figure 5: Change in Real Land Values per Acre Around the Coastal Zone Boundary Pre- and Post-2016 Reforms



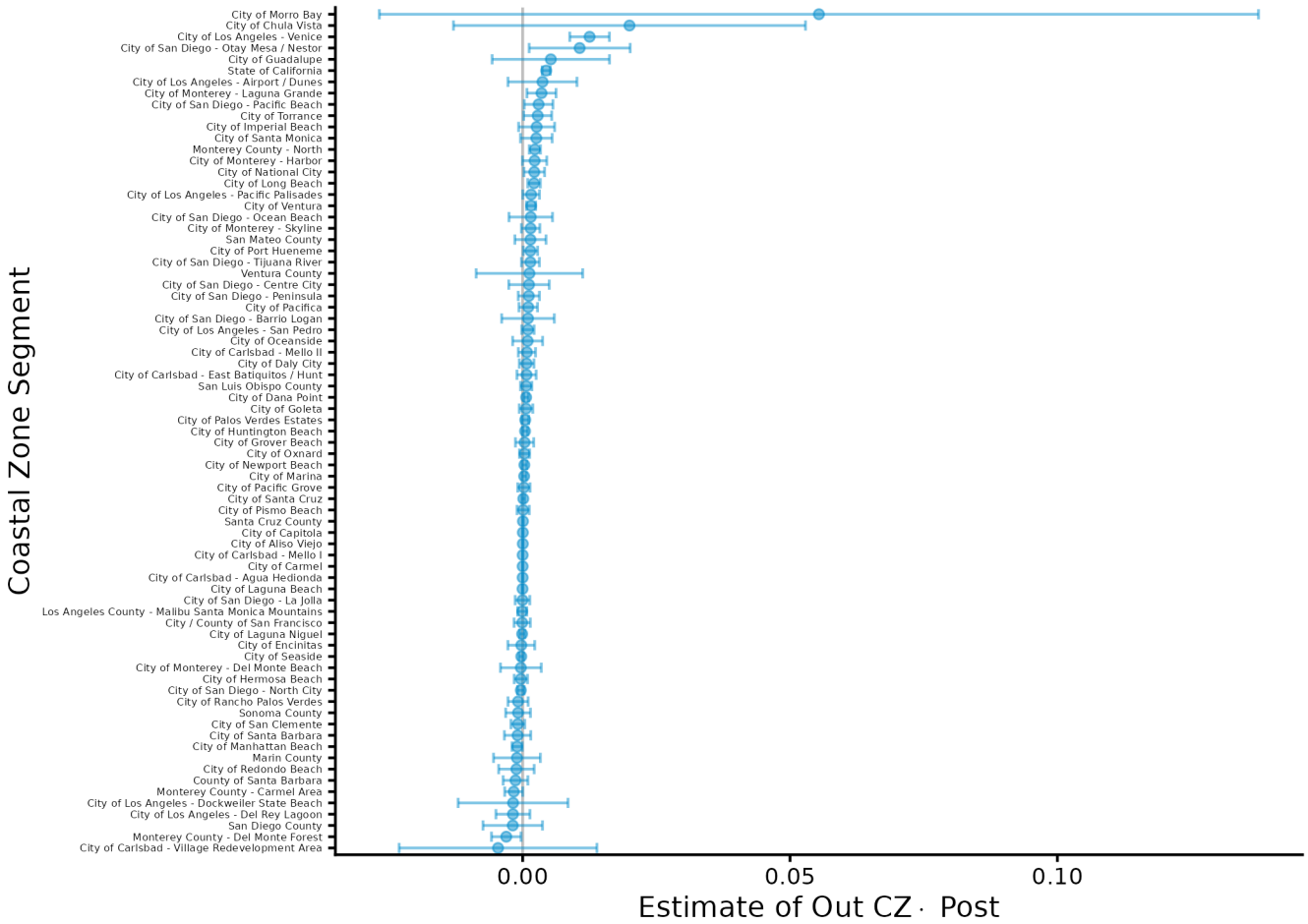
This figure uses data from Cotality (2025) and California Coastal Commission (2014) to show how residential land values change across the Coastal Zone boundary between 2010 and 2025. The horizontal axis shows each bin's distance from the Coastal Zone boundary, where negative values represent parcels inside the Coastal Zone that do not receive the post 2016 permitting reforms. We group parcels in 50 meter bins. The vertical axis shows the log of the real land value per acre for each bin. The orange squares show the 2010-2016 distribution of land values in relation to the Coastal Zone boundary. The blue circles show the 2017 to 2025 distribution of land values in relationship to the the Coastal Zone boundary. The difference in discontinuities estimate can be seen in comparing the gap between the pre- and post-2016 lines just before and then just after crossing the Coastal Zone boundary at 0.

Figure 6: Real Total Sale Value Per Acre Changes By Coastal Zone Distance Bin



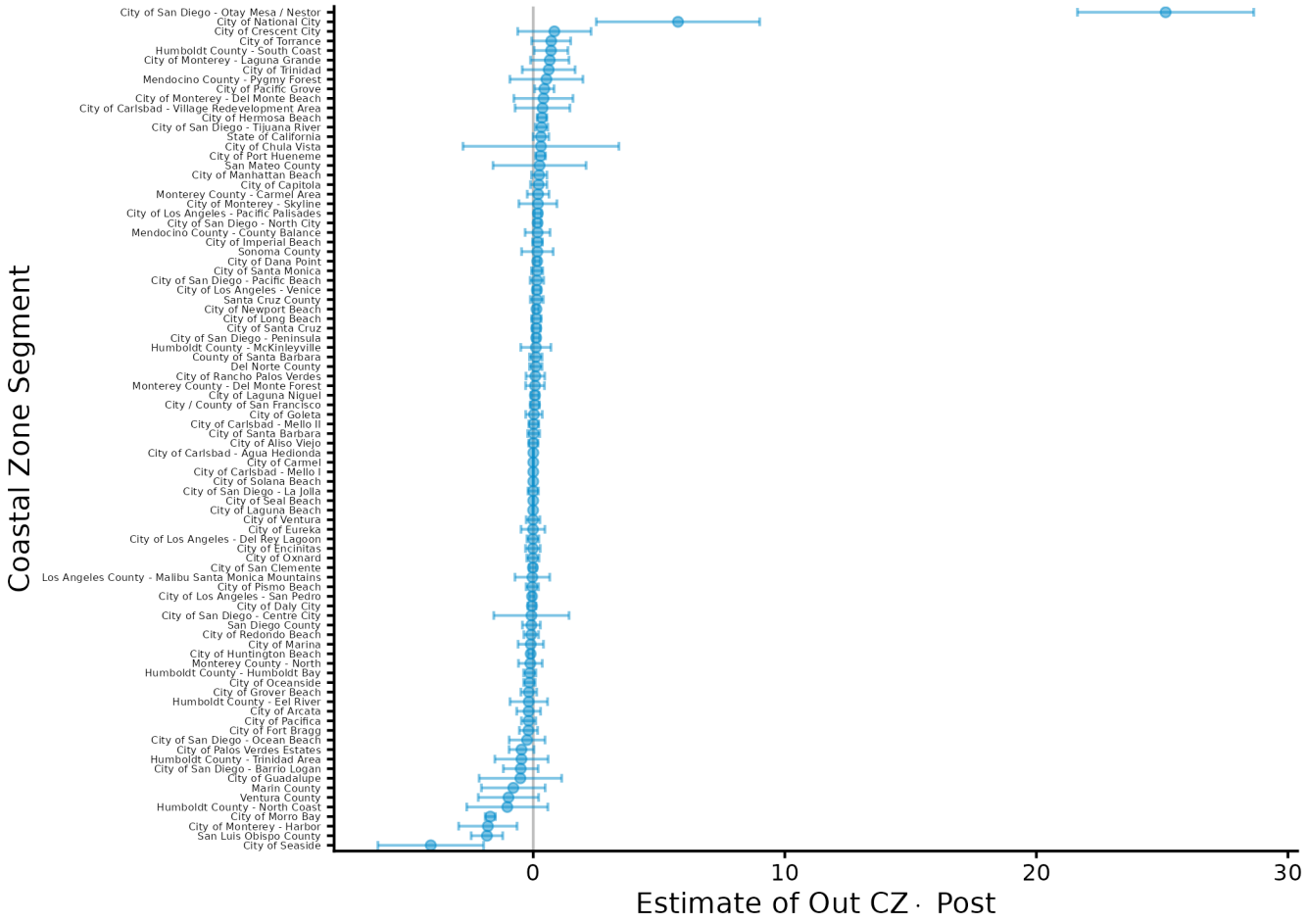
This figure uses data from Cotality (2025) and California Coastal Commission (2014) to show how changes in the total sale price per acre change across the Coastal Zone boundary between 2010 and 2016. The horizontal axis shows each bin's distance from the Coastal Zone boundary, where negative values represent parcels inside the Coastal Zone that do not receive the post 2016 permitting reforms. We group parcels in 50 meter bins. The vertical axis shows the coefficients from Equation 3 as described in the text. These coefficients are from a regression with the log of the real total sale price per acre as the dependent variable. These coefficients are on the set of distance bin times post interactions, where a single bin has been omitted to serve as the reference bin. The coefficients can be interpreted as percentage changes relative to the omitted bin. So, the first coefficient outside the Coastal Zone ($[0, 50)$) saw transaction prices increase by over 5% more than transactions in the reference bin ($[-500, -450)$) between 2017 and 2025 compared to the 2010 to 2016 baseline.

Figure 7: Treatment Effect Differences in Accessory Dwelling Unit Permitting Across Coastal Zone Segments



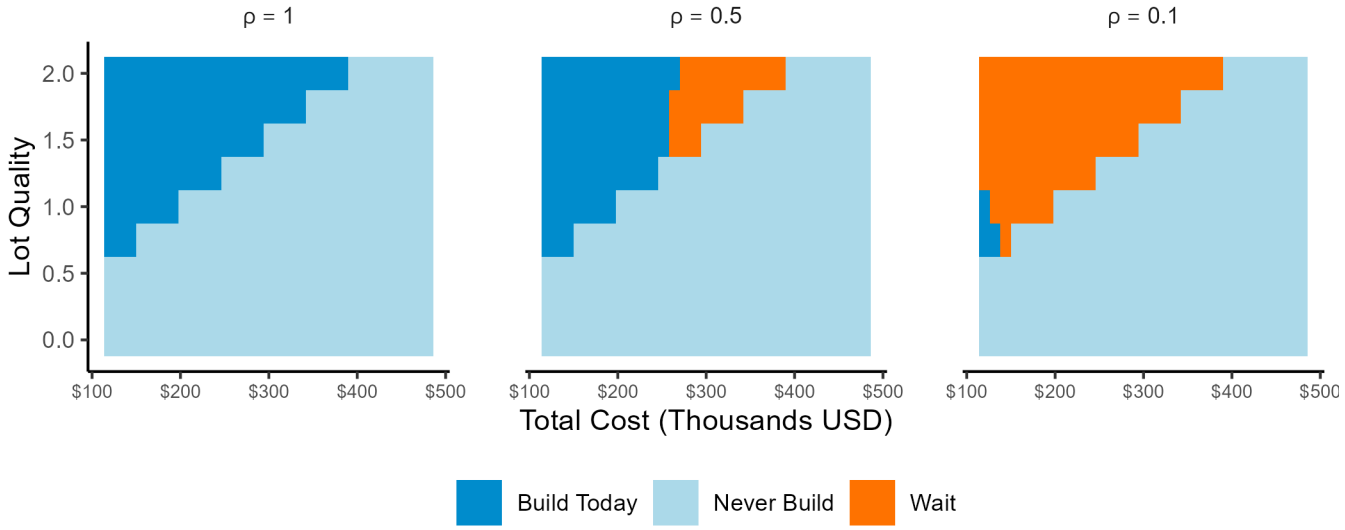
This figure uses data from Cotality (2025), California Department of Housing and Community Development (2025), and California Coastal Commission (2014) to estimate a modified version of Equation 1 for each Coastal Zone segment separately. The outcome variable is the dummy for whether or not a parcel has an ADU in a given year. The regressions include a first degree polynomial of distance from the Coastal Zone boundary, a set of county fixed effects, and a set of year fixed effects. The coefficients are on the dummy for a parcel being outside of the Coastal Zone and therefore receiving the post-2016 permitting reforms for ADUs. Robust standard errors are clustered at the parcel level. Note that there are several segments of the Coastal Zone where all parcels are located either completely within the Coastal Zone or completely outside the Coastal Zone. These segments cannot be included in this exercise.

Figure 8: Treatment Effect Differences in Real Land Values Per Acre Across Coastal Zone Segments



This figure uses data from Cotality (2025) and California Coastal Commission (2014) to estimate a modified version of Equation 2 for each Coastal Zone segment separately. The outcome variable is the log of the real land value per acre for transaction. The regressions include a first degree polynomial of distance from the Coastal Zone boundary, a set of county fixed effects, and a set of year fixed effects. The coefficients are on the dummy for a parcel being outside of the Coastal Zone and interacted with the post 2016 dummy variable. Areas outside the Coastal Zone receive the post-2016 permitting reforms for ADUs. Robust standard errors are clustered at the parcel level. Note that there are several segments of the Coastal Zone where all parcels are located either completely within the Coastal Zone or completely outside the Coastal Zone. These segments cannot be included in this exercise.

Figure 9: Model Solutions by Land Quality and Total Net Cost to Build an ADU



This figure shows solutions to the model presented in Sub-section 6.3. The model is solved for 10 years. Interest raters are set at 10% (i) and rents from existing structures (s) are set at $50,000 per year. We use a 1,200 square foot ADU that rents for $1,200 per month in the base period, $2,400 with high rents, and -$1,200 with low rents for a base land quality l. We calculate solutions for all combinations of $l \in \{0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2\}$, $c \in \{100, 150, 200, 250, 300, 350, 400, 450, 500\}$ with c in thousands of USD, and $\rho \in \{0.1, 0.5, 1\}$. These build costs reflect industry ranges from Henson (2023).

Table 1: Sample Statistics for the Transaction Data by Coastal Zone Sample Status and Reform Period

	In CZ		Out CZ		Full County	
	Pre-2016	Post-2016	Pre-2016	Post-2016	Pre-2016	Post-2016
	(1)	(2)	(3)	(4)	(5)	(6)
Share SFR	0.882	0.895	0.899	0.904	0.918	0.927
	(0.323)	(0.306)	(0.302)	(0.295)	(0.274)	(0.26)
Dist. to the CZ	-429.82	-433.06	469.852	468.478	15,584	15,498
	(275.654)	(275.099)	(284.303)	(284.626)	(16,013)	(16,085)
Effect. Build Yr.	1,968	1,972	1,966	1,969	1,965	1,969
	(23.937)	(24.721)	(24.394)	(24.16)	(23.438)	(25.811)
Bedrooms	3.267	3.304	3.297	3.352	3.33	3.372
	(2.103)	(1.276)	(1.46)	(4.377)	(1.97)	(1.952)
Bathrooms	2.552	2.753	2.408	2.533	2.341	2.463
	(1.239)	(1.355)	(1.232)	(1.253)	(1.202)	(1.21)
Building Sq. Ft.	2,007	2,222	1,896	2,051	1,877	2,022
	(1,073)	(1,284)	(1,148)	(1,260)	(2,455)	(2,019)
Land Val. / Acre	7,017,159	9,361,767	4,894,626	7,029,095	3,130,824	4,521,749
	(6,886,149)	(8,109,829)	(4,775,258)	(6,486,825)	(3,747,236)	(4,911,504)
Tot. Sale Pr. / Acre	11,120,235	14,615,388	8,238,706	11,500,262	5,627,400	7,957,786
	(9,193,994)	(10,739,012)	(6,985,092)	(9,064,985)	(5,771,560)	(7,270,414)
Land Share	0.589	0.599	0.563	0.573	0.529	0.531
	(0.195)	(0.198)	(0.186)	(0.195)	(0.191)	(0.197)

Notes: This table shows summary statistics for the transaction dataset. Standard deviations are shown in parentheses. Columns (1)-(2) show parcels within the Coastal Zone, columns (3)-(4) show parcels outside the Coastal Zone but within 1,000 meters of the boundary, and columns (5)-(6) show all parcels in the 15 coastal counties of California. We present means and standard deviations for the share of parcels on land zoned only for single family residences, the parcel's distance to the Coastal Zone in meters, the effective build year of the structure(s) on the parcel, the number of bedrooms, number of bathrooms, building square footage, the real land value per acre, the real total sale price per acre, and the land share of the total value. Dollar values are in 2025 USD.

Table 2: Sample Statistics for the Balanced Parcel Panel Data by Coastal Zone Sample Status and Reform Period

	In CZ		Out CZ		Full County	
	Pre-2016	Post-2016	Pre-2016	Post-2016	Pre-2016	Post-2016
	(1)	(2)	(3)	(4)	(5)	(6)
Share SFR	0.861	0.87	0.893	0.901	0.908	0.914
	(0.345)	(0.336)	(0.309)	(0.299)	(0.289)	(0.28)
Dist. to the CZ	-432.098	-432.196	463.349	463.346	14,486	14,485
	(275.13)	(275.148)	(284.446)	(284.444)	(14,750)	(14,749)
Effect. Build Yr.	1,967	1,971	1,965	1,970	1,964	1,967
	(23.265)	(24.503)	(22.902)	(23.406)	(23.521)	(24.617)
Bedrooms	3.282	3.348	3.336	3.386	3.364	3.426
	(1.494)	(1.759)	(4.108)	(2.621)	(2.256)	(2.375)
Bathrooms	2.554	2.738	2.367	2.478	2.33	2.432
	(1.895)	(2.743)	(2.925)	(1.294)	(1.647)	(1.401)
Building Sq. Ft.	2,009	2,196	1,913	2,055	1,929	2,057
	(1,775)	(2,194)	(2,668)	(2,844)	(43,623)	(43,093)
Land Val. / Acre	8,702,655	13,009,492	4,922,152	7,520,758	3,515,259	5,453,182
	(13,514,125)	(21,427,760)	(7,939,038)	(11,497,427)	(110,387,961)	(54,585,213)
Tot. Sale Pr. / Acre	13,405,476	19,474,269	8,936,441	13,212,545	6,956,994	10,394,807
	(19,383,560)	(33,433,248)	(23,680,285)	(38,842,937)	(139,231,439)	(130,230,547)
Land Share	0.552	0.578	0.521	0.55	0.503	0.532
	(0.207)	(0.211)	(0.193)	(0.199)	(0.19)	(0.195)
ADU Permit		0.001		0.002		0.002
		(0.034)		(0.04)		(0.048)
Any Permit		0.002		0.002		0.003
		(0.046)		(0.047)		(0.058)

Notes: This table shows summary statistics for the balanced parcel panel. Standard deviations are shown in parentheses. Columns (1)-(2) show parcels within the Coastal Zone, columns (3)-(4) show parcels outside the Coastal Zone but within 1,000 meters of the boundary, and columns (5)-(6) show all parcels in the 15 coastal counties of California. We present means and standard deviations for the share of parcels on land zoned only for single family residences, the parcel's distance to the Coastal Zone in meters, the effective build year of the structure(s) on the parcel, the number of bedrooms, number of bathrooms, building square footage, the real land value per acre, the real total sale price per acre, the land share of the total value, whether or not the parcel is associated with an accessory dwelling unit permit, and whether or not the parcel is associated with any building permit. Dollar values are in 2025 USD.

Table 3: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Accessory Dwelling Unit Permits after 2018 within 1,000 Meters of the Coastal Zone Boundary

	Dep.Var.: Parcel has ADU Permit within the Year							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
OutCZ	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
1st Degree Polynomial	Y		Y		Y		Y	
2nd Degree Polynomial		Y		Y		Y		Y
County FEs	Y	Y	Y	Y	Y	Y	Y	Y
Year FEs	Y	Y	Y	Y				
Segment · Time FEs					Y	Y	Y	Y
Pre-2016 Static Hedonics			Y	Y			Y	Y
N	3 291 136	3 291 136	3 247 329	3 247 329	3 291 120	3 291 120	3 247 321	3 247 321
Adj. R^2	0.001	0.001	0.002	0.002	0.003	0.003	0.003	0.003
Mean	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002
SD	0.041	0.041	0.041	0.041	0.041	0.041	0.041	0.041
In-CZ Mean	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
In-CZ SD	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036

Notes: This table uses data from Cotality (2025), California Department of Housing and Community Development (2025), and California Coastal Commission (2014) to estimate Equation 1 for a balanced set of residential parcels between 2018 and 2025: $Y_{ict} = \alpha + \beta_1 OutCZ_i + \beta_2 f(DistCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable is a dummy taking a value of 1 if parcel i has an ADU permit associated with it in year t. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's permitting reforms for ADUs. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first degree or second degree polynomial form. In both polynomial specifications, distance slopes can change after the boundary. We also include a set of county fixed effects, λ_c, and a set of year fixed effects, τ_t. We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We cluster our robust standard errors at the parcel level. We report estimates using a 1,000 meter bandwidth from the Coastal Zone boundary. We present summary statistics for the permit dummy for both the entire sample and just the sample within the coastal zone at the bottom of the table. In Column 1, we present a specification only with a first degree polynomial, a set of county fixed effects, and a set of year fixed effects. In the even columns, we replace the first degree polynomial with the second degree polynomial. Then in Column 3 and 4, we include the set of hedonic fixed effects. Then in columns 5 through 8, we replace the year fixed effects with a segment by time fixed effect that limits comparisons to only be within each parcel's closest segment of the Coastal Zone boundary. *$p < .05$, ** $p < .01$, *** $p < .001$

Table 4: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Land Values within 1,000 Meters of the Coastal Zone Boundary

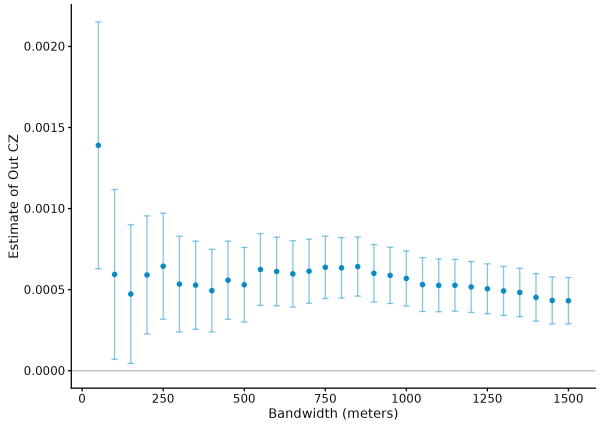
	Dep. Var.: Log of Real Land Value Per Acre							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$OutCZ \cdot Post$	0.103***	0.104***	0.092***	0.095***	0.076***	0.066**	0.069***	0.059**
	(0.017)	(0.025)	(0.017)	(0.025)	(0.015)	(0.022)	(0.015)	(0.022)
1st Degree Polynomial	Y		Y		Y		Y	
2nd Degree Polynomial		Y		Y		Y		Y
County FEs	Y	Y	Y	Y	Y	Y	Y	Y
Year FEs	Y	Y	Y	Y				
Segment · Time FEs					Y	Y	Y	Y
Pre-2016 Static Hedonics			Y	Y			Y	Y
N	200 687	200 687	198 812	198 812	200 659	200 659	198 784	198 784
Adj. R^2.	0.496	0.496	0.518	0.519	0.618	0.618	0.635	0.635
Mean Pre-2016	15	15	15	15	15	15	15	15
SD Pre-2016	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
In-CZ Mean Pre-2016	15.2	15.2	15.2	15.2	15.2	15.2	15.2	15.2
In-CZ SD Pre-2016	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3

Notes: This table uses data from Cotality (2025) and California Coastal Commission (2014) to estimate Equation 2 for a transactions on residential parcels between 2010 and 2025: $ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3 f(DistCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable the log of the real land value per acre in 2025 USD. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 after 2016 demarcating the implementation of the first ADU reforms. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first degree or second degree polynomial form. In both polynomial specifications, distance slopes can change after the boundary. We also include a set of county fixed effects, λ_c, and a set of year fixed effects, τ_t. We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We cluster our robust standard errors at the parcel level. We report estimates using a 1,000 meter bandwidth from the Coastal Zone boundary. We present summary statistics for the log of the real land value per acre for both the entire sample from 2010 to 2016 and just the sample within the coastal zone from 2010 to 2016 at the bottom of the table. In Column 1, we present a specification only with a first degree polynomial, a set of county fixed effects, and a set of year fixed effects. In the even columns, we replace the first degree polynomial with the second degree polynomial. Then in Column 3 and 4, we include the set of hedonic fixed effects. Then in columns 5 through 8, we replace the year fixed effects with a segment by time fixed effect that limits comparisons to only be within each parcel's closest segment of the Coastal Zone boundary. *$p < .05$, ** $p < .01$, *** $p < .001$

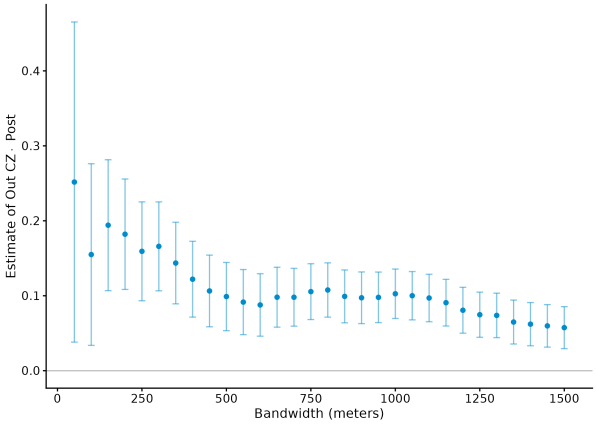
A Additional Exhibits

Figure A.1: Robustness of Accessory Dwelling Permitting and Real Land Value Per Acre Estimates to Bandwidth Choice

(a) Panel A: Accessory Dwelling Unit Permits

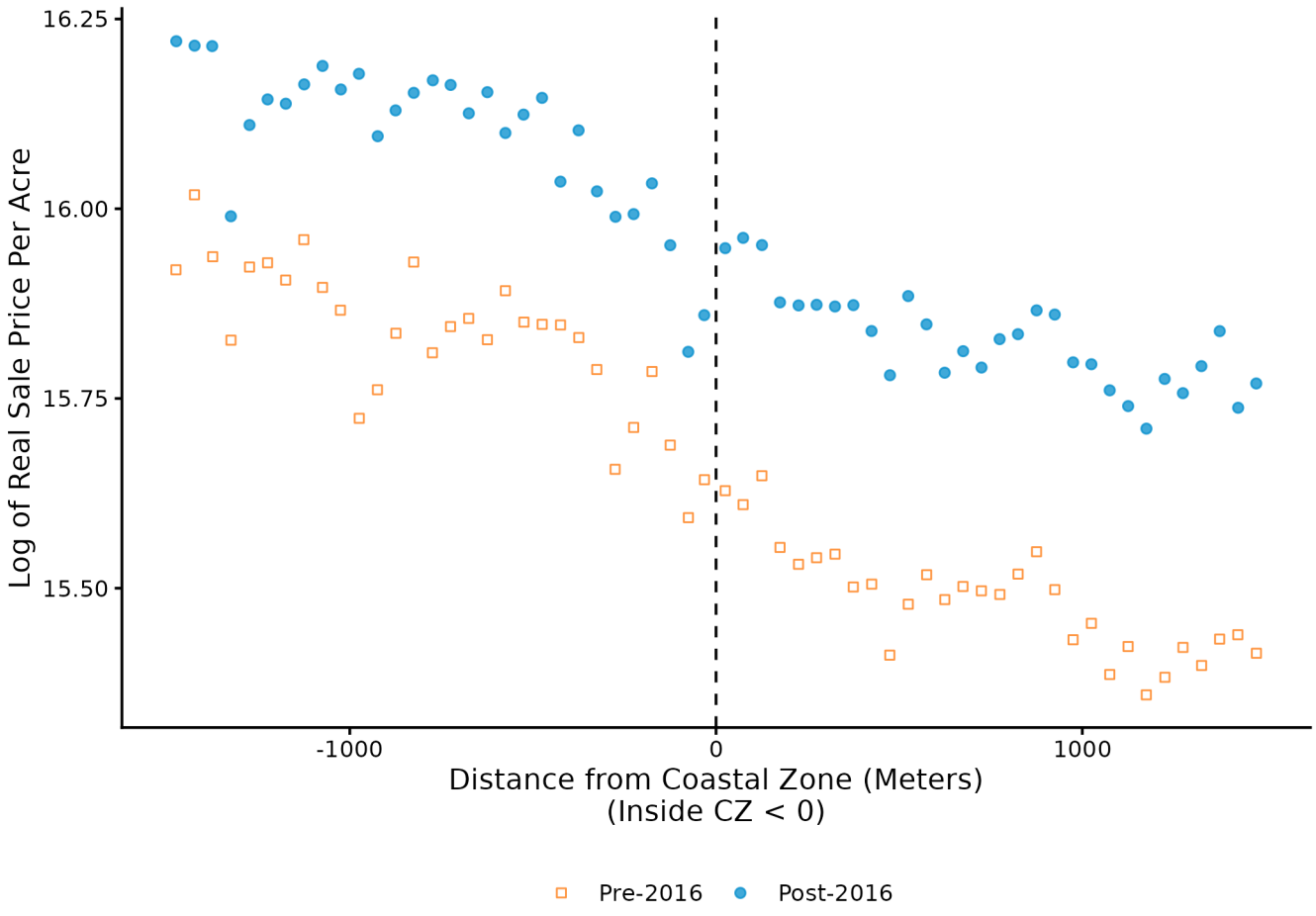


(b) Panel B: Real Land Values Per Acre



This figure shows estimates of Equation 1 in Panel A and 2 in Panel B for alternative choices for the sample bandwidth cutoff. We show these (symmetric) sample bandwidths on the horizontal axis and the coefficients of interest on the vertical axis. These estimates correspond to Column 1 in Table 3 in Panel A and Column 1 in Table 4 in Panel B. The outcome variable in Panel A is a dummy taking a value of 1 is a parcel gets a permit for an ADU in a particular year and 0 otherwise. The outcome variable in Panel B is the log of the real land value per acre in 2025 USD. We use our simple baseline specification with a first degree polynomial of distance, a set of county fixed effects, and a set of year fixed effects. Robust standard errors are clustered at the parcel level in both panels. Our chose bandwidth for the main estimates is 1,000 meters from the Coastal Zone boundary.

Figure A.2: Change in Real Total Sale Prices per Acre Around the Coastal Zone Boundary Pre- and Post- 2016 Reforms



This figure uses data from Cotality (2025) and California Coastal Commission (2014) to show how residential total sale values change across the Coastal Zone boundary between 2010 and 2025. The horizontal axis shows each bin's distance from the Coastal Zone boundary, where negative values represent parcels inside the Coastal Zone that do not receive the post 2016 permitting reforms. We group parcels in 50 meter bins. The vertical axis shows the log of the real total sale price per acre for each bin. The orange squares show the 2010-2016 distribution of total sale values in relation to the Coastal Zone boundary. The blue circles show the 2017 to 2025 distribution of total sale values in relationship to the the Coastal Zone boundary. The difference in discontinuities estimate can be seen in comparing the gap between the pre- and post-2016 lines just before and then just after crossing the Coastal Zone boundary at 0.

Table A.1: Summary Statistics for the Full County Sample

Panel A: Transaction Data	Mean	SD	Min	Max
	(1)	(2)	(3)	(4)
Share SFR	0.923	0.267	0	1
Dist. to the CZ	15,540	16,049	-42,120	103,660
Effective Build Year	1,967	24.829	25	8,600
Bedrooms	3.352	1.961	1	999
Bathrooms	2.404	1.208	1	224
Building Sq. Ft.	1,954	2,235	1	1,320,623
Land Value Per Acre	3,832,895	4,428,751	7.878	49,570,267
Total Sale Price Per Acre	6,803,615	6,673,447	20,510	68,344,349
Land Share	0.53	0.194	0	1

Panel B: Parcel Panel Data	Mean	SD	Min	Max
	(1)	(2)	(3)	(4)
Share SFR	0.911	0.284	0	1
Dist. to the CZ	14,485	14,749	-44,393	108,132
Effective Build Year	1,966	24.253	3	8,600
Bedrooms	3.4	2.325	1	999
Bathrooms	2.388	1.511	1	999
Building Sq. Ft.	2,004	43,313	0	77,952,576
Land Value Per Acre	4,465,389	87,595,185	0	102,076,318,951
Total Sale Price Per Acre	8,642,407	134,904,672	0.058	128,726,779,453
Land Share	0.519	0.193	0	1
ADU Permit	0.001	0.036	0	1
Any Permit	0.002	0.043	0	1

Notes: This table presents summary statistics for the full 15 coastal county samples using data from Cotality (2025), California Coastal Commission (2014), and California Department of Housing and Community Development (2025). We show summary statistics using the transaction dataset in Panel A, which uses transaction records from 2010 to 2025. In Panel B we show summary statistics for the balanced panel used for evaluating permitting changes from 2018 to 2025. We show summary statistics for the share of parcels that are zoned for single family use, the parcel's euclidean distance to the Coastal Zone boundary, the parcel's effective build year, the parcel structure's number of bedroom, bathrooms, building square footage, land values per acre, total sale value per acre, and the share of value assessed to come from the land. In Panel B, we also show the share of parcels in a particular year that have an ADU permit or any building permit associated with them.

Table A.2: Cumulative Description of Sample Restrictions

Sample Restriction	Transaction Data (1)	Balanced Panel (2)
Total Observations	10,067,738	82,360,928
+ Residential Properties	8,117,520	66,608,560
+ Arms-Length Transactions	2,368,275	–
+ Assessed Value $\geq$ \$1,000	2,368,230	66,580,309
+ Valid Sale Price	2,328,496	–
+ Sale Count Not Extreme	2,275,883	–
+ Valid Rooms/Units Ratio	2,275,883	66,482,873
+ Land Value Per Acre > \$1	2,161,709	–
+ Valid Implied Age	2,161,709	66,482,792
Final Sample	2,161,709	66,482,792

Notes: This table presents the impact of sample restrictions for the full 15 coastal county samples using data from Cotality (2025), California Coastal Commission (2014), and California Department of Housing and Community Development (2025). We show how each restriction reduces the total number of observations cumulatively descending through the rows. To start, all parcels must have an identifier and a valid location (latitude and longitude). Column 1 shows how restrictions impact the transaction dataset. These include a restriction to small residential properties, generally single family houses but strictly just excluding large multifamily units and condos; restricting to arms-length (non family) transactions; limiting to parcels with assessed values greater than \$1000 USD; having a sale price within the 1st and 99th percentile of the sample distribution; not having an extreme number of transactions above the 99th percentile of the sample distribution per year; having a room to unit ratio within the 1st and 99th percentile of the sample distribution; having a land value per acre above \$1; and having an internally consistent structure age. In Column 2, we show these restrictions for the balanced panel dataset, which has one observation per parcel between 2010 and 2025, when the restriction is applicable since the data is only based on assessment records and not transaction records.

Table A.3: Interaction of Sample Restrictions with the Coastal Zone Boundary in the Transaction and Parcel Datasets

	Transactions (1)	Parcel Panel (2)
$OutCZ \cdot Post$	0.000	
	(0.000)	
$OutCZ$		-0.035***
		(0.002)
1st Degree Polynomial	Y	Y
County FEs	Y	Y
Year FEs	Y	Y
N.	9 355 011	4 442 024
R2 Adj.	0.009	0.086
Years	2010 - 2025	2018 - 2025

Notes: The following table uses data from Cotality (2025) and California Department of Housing and Community Development (2025) to see how sample restrictions interact with our baseline regression equations. In both columns the outcome variable is a dummy variable for inclusion in the analysis sample, taking a value of 1 if an observation is trimmed from the final sample. In Column 1, we show the baseline version of Equation 2, which shows us that our sample exclusions are not changing over time. In Column 2, we show our baseline version of Equation 1 for only the 2018 and after sample in which we run our regressions. Here, we see that properties inside the Coastal Zone are more likely to be excluded. However, from Column 1 we can see that these restrictions likely don't change over time. *$p < .05$, ** $p < .01$, *** $p < .001$

Table A.4: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Land Values within 1,000 Meters of the Coastal Zone Boundary Using Parcels with Repeat Sales

	Dep. Var.: Log of Real Land Value Per Acre			
	(1)	(2)	(3)	(4)
$OutCZ \cdot Post$	0.039**	0.075***	0.042**	0.072***
	(0.014)	(0.020)	(0.014)	(0.020)
Parcel FEs	Y	Y	Y	Y
Year FEs	Y	Y		
Segment · Time FEs			Y	Y
1st Degree Polynomial	Y		Y	
2nd Degree Polynomial		Y		Y
N	95 288	95 288	95 210	95 210
Adj. R^2	0.900	0.900	0.903	0.903
Mean Pre-2016	15	15	15	15
SD Pre-2016	1.2	1.2	1.2	1.2
In-CZ Mean Pre-2016	15.2	15.2	15.2	15.2
In-CZ SD Pre-2016	1.3	1.3	1.3	1.3

Notes: This table uses data from Cotality (2025) and California Coastal Commission (2014) to estimate Equation 2 for a transactions on residential parcels with repeat sales between 2010 and 2025: $ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3 f(DistCZ_i) + \psi_i + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. To be considered a repeat sale parcel, it must be part of a transaction between both 2010 to 2016 and 2017 to 2025. The outcome variable the log of the real land value per acre in 2025 USD. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 after 2016 demarking the implementation of the first ADU reforms. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first degree or second degree polynomial form. In both polynomial specifications, distance slopes can change after the boundary. We also include a set of parcel fixed effects, ψ_i, and a set of year fixed effects, τ_t. ε_{ict} is an error term. We cluster our robust standard errors at the parcel level. We report estimates using a 1,000 meter bandwidth from the Coastal Zone boundary. We present summary statistics for the log of the real land value per acre for both the entire sample from 2010 to 2016 and just the sample within the coastal zone from 2010 to 2016 at the bottom of the table. In Column 1, we present a specification only with a first degree polynomial, a set of parcel fixed effects, and a set of year fixed effects. In the even columns, we replace the first degree polynomial with the second degree polynomial. Then in Column 3 and 4, we replace the year fixed effects with a segment by time fixed effect that limits comparisons to only be within each parcel's closest segment of the Coastal Zone boundary. *$p < .05$, ** $p < .01$, *** $p < .001$

Table A.5: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Land Values within 1,000 Meters of the Coastal Zone Boundary Using Time-Varying Hedonics

	Dep. Var.: log of Real Sale Price Per Acre			
	(1)	(2)	(3)	(4)
$OutCZ \cdot Post$	0.094***	0.086***	0.076***	0.054**
	(0.015)	(0.022)	(0.013)	(0.019)
1st Degree Polynomial	Y		Y	
2nd Degree Polynomial		Y		Y
County FEs	Y	Y	Y	Y
Year FEs	Y	Y		
Segment · Time FEs			Y	Y
Updating Hedonics	Y	Y	Y	Y
N	200 662	200 662	200 634	200 634
Adj. R^2	0.493	0.494	0.619	0.619
Mean Pre-2016	15.6	15.6	15.6	15.6
SD Pre-2016	1.1	1.1	1.1	1.1
In-CZ Mean Pre-2016	15.8	15.8	15.8	15.8
In-CZ SD Pre-2016	1.1	1.1	1.1	1.1

Notes: This table uses data from Cotality (2025) and California Coastal Commission (2014) to estimate Equation 2 for a transactions on residential parcels between 2010 and 2025: $ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3 f(DistCZ_i) + \theta'_{(it)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable the log of the real total sale price per acre in 2025 USD. In each specification, we use a set of binned time varying hedonic fixed effects, $\theta'_{(it)}$, to capture the share of the total value associated with the market values of the current structural characteristics of the parcel. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 after 2016 demarking the implementation of the first ADU reforms. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first degree or second degree polynomial form. In both polynomial specifications, distance slopes can change after the boundary. We also include a set of county fixed effects, λ_c, and a set of year fixed effects, τ_t. ε_{ict} is an error term. We cluster our robust standard errors at the parcel level. We report estimates using a 1,000 meter bandwidth from the Coastal Zone boundary. We present summary statistics for the log of the real total sale price per acre for both the entire sample from 2010 to 2016 and just the sample within the coastal zone from 2010 to 2016 at the bottom of the table. In Column 1, we present a specification only with a first degree polynomial, the set of time varying hedonic fixed effects, a set of county fixed effects, and a set of year fixed effects. In the even columns, we replace the first degree polynomial with the second degree polynomial. Then in Column 3 and 4, we replace the year fixed effects with a segment by time fixed effect that limits comparisons to only be within each parcel's closest segment of the Coastal Zone boundary. *$p < .05$, ** $p < .01$, *** $p < .001$